

Hypoxic Blackout in Serial Freediving – Protective Mechanisms and Risk Factors

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To Cristina and Valentina, the loves of my life

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Abstract

Breath-hold diving or freediving exposes the body to stressors such as low oxygen and high carbon dioxide levels in the blood, increased hydrostatic pressure, and cold. The reduction in blood oxygen levels is considered a primary factor for loss of consciousness while diving. This is known as hypoxic blackout. My thesis aimed to enhance the understanding of the factors that lead to hypoxic blackout in serial freediving. This includes studying the physiological mechanisms that protect the body against hypoxia and exploring how hyperventilation affects those mechanisms, with a specific focus on repetitive dives.

The protective mechanisms, splenic contraction (Studies I, II) and diving bradycardia (Studies II and IV), were tested in experimental conditions. The effect of hyperventilation on simulated serial dives (Study II) was investigated, and a strategy to identify actual hyperventilation during diving was developed and explored (Study III). A static apnea test allowing identification of individuals at particular risk for blackout was developed (Study IV).

The results showed that hypoxia-induced splenic contraction occurs rapidly enough to be protective in long-duration breath-holds (Study I). Serial simulated dives, preceded by short-term hyperventilation, lead to longer apnea duration and progressive oxygen desaturation (Study II). Despite the more severe desaturation resulting from hyperventilation, these consecutive apneas do not show an augmented diving response or splenic contraction. Hyperventilation was also observed when divers intended to avoid it (Studies II, III, and IV). Therefore, the possibility of estimating hyperventilation from the signal of a force sensor attached to a chest belt was explored (Study III). A stronger diving bradycardia and a bigger spleen were found to be protective against hypoxia and are related to slower oxygen desaturation (Study IV).

The main conclusion is that oxygenation is impaired by the accumulation of an oxygen debt when consecutive dives are preceded by hyperventilation. The protective mechanisms, such as

splenic contraction and the diving response, are beneficial at the individual level but do not offset the increased risk across a series of dives. However, splenic contraction does offer protection even during the first dive of a series. A chest belt-mounted force sensor could be used to identify when freedivers are hyperventilating. Additionally, a novel static apnea ramp test is effective in identifying freedivers who are at a high risk of faster desaturation during freediving.

Keywords: breath-hold diving, diving response, hyperventilation, hypoxia, oxygen saturation, oxygen stores, pulse oximetry, sonography, splenic contraction.

Summary in Swedish

Fridykning utsätter kroppen för fysisk stress så som låga syrenivåer och höga koldioxidnivåer i blodet, ökat hydrostatiskt tryck och kyla. Minskningen av syrenivåerna i blodet anses vara en primär orsak till medvetlöshet under dykning. Detta kallas hypoxisk blackout. Min avhandling syftar till att öka kunskapen om de faktorer som leder till hypoxisk blackout vid upprepade fridykning. Detta inkluderar studier av de fysiologiska mekanismerna som skyddar kroppen mot hypoxi och utforskande av hur hyperventilation påverkar dessa mekanismer, med speciellt fokus på upprepade dyk.

De skyddande mekanismerna, mjältkontraktion (Studierna I, II) och dykbradykardi (Studierna II och IV), testades under experimentella förhållanden. Effekten av hyperventilation på simulerade seriella dyk (Studie II) undersöktes och en strategi för att identifiera faktisk hyperventilation under dykning utvecklades och utforskades (Studie III). Ett statiskt apnétest som möjliggör identifiering av individer med särskild risk att drabbas av blackout utvecklades (Studie IV).

Resultaten visade att hypoxi-inducerad mjältkontraktion sker tillräckligt snabbt för att vara skyddande vid långa andhållningar (Studie I). Serier av simulerade dyk, som föregås av kortvarig hyperventilering, leder till längre apnéer och progressiv sänkning av syremättnaden i blodet (studie II). Trots den minskade syremättnaden som blev resultatet av hyperventilering vid seriella apnéer, kompenserades detta inte av en ökad dykrespons eller mjältsammandragning. Hyperventilering observerades också när dykarna hade för avsikt att undvika det (studier II, III och IV). Därför undersöktes möjligheten att uppskatta hyperventilation från signalen från en kraftsensor fäst vid ett bröstbälte (Studie III). På individnivå visade sig en starkare dykbradykardi och en större mjälte vara skyddande mot hypoxi och relaterade till långsammare desaturation (Studie IV).

Huvudslutsatsen är att syretillgången försämras genom en ackumulerad syreskuld när seriella dyk föregås av hyperventilering.

De skyddande mekanismerna, såsom mjältkontraktion och dykresponsen, är fördelaktiga på individnivå men uppväger inte den ökade risken vid seriella dyk. Mjälteskontraktion ger dock skydd även under det första dyket i en serie. En kraftsensor som är monterad på ett bröstbälte kan användas för att identifiera när fridykare hyperventilerar. Dessutom visades det att ett nytt ramptest baserat på statisk apné är effektivt för att identifiera fridykare som löper hög risk för en snabb desaturation under fridykning.

Nyckelord: andhållningsdykning, dykrespons, hyperventilering, hypoxi, syresättning, syreförråd, pulsoximetri, sonografi, mjälteskontraktion.

List of papers

This thesis is based on the following four papers, herein referred to by their Roman numerals:

- Study I Pernett F., Schagatay F., Vildevi C., Schagatay E. Spleen Contraction During Sudden Eupneic Hypoxia Elevates Hemoglobin Concentration. *Frontiers in Physiology*, 2021 Sep 21;12:729123. doi: 10.3389/fphys.2021.729123.
- Study II Pernett F., Bergenhed P., Holmström P., Mulder E., Schagatay E. Effects of hyperventilation on oxygenation, apnea breaking points, diving response, and spleen contraction during serial static apneas. *European Journal of Applied Physiology*. 2023 Aug;123(8):1809-1824. doi: 10.1007/s00421-023-05202-7.
- Study III Pernett F., Mulder E., Johansson F., Sieber A., Bermudez R., Lossner M., Schagatay E. Estimating End-tidal Carbon Dioxide in Freedivers Using an Underwater Force Sensor. *European Journal of Applied Physiology. In review*
- Study IV Pernett F., Schagatay F., Löfquist I., Johansson F., Schagatay E. Implementation of a static apnea ramp test to evaluate the risk of severe oxygen desaturation in freedivers. *In manuscript*

Abbreviations

η_p^2	Partial eta squared
η_G^2	Generalized eta squared
AIDA	International Association for the development of apnea
ANOVA	Analysis of variance
BMI	Body mass index
BO	Blackout
CMAS	World Federation of Underwater Activities
CaO_2	Arterial oxygen content
DAN	Divers Alert Network
$\dot{V}O_2$	Oxygen delivery
ES	Effect size
FiO_2	Inspired fraction of oxygen
H^+	Hydrogen ion
Hb	Hemoglobin concentration
HCO_3	Bicarbonate
HR	Heart rate
IBM	Involuntary breathing movements
ITBV	Intra-thoracic blood volume
MAP	Mean arterial blood pressure

$P_a\text{CO}_2$	Arterial pressure of carbon dioxide
$P_A\text{O}_2$	Alveolar pressure of oxygen
$P_a\text{O}_2$	Arterial pressure of oxygen
$P_{\text{ET}}\text{CO}_2$	End-tidal pressure of carbon dioxide
$P_{\text{ET}}\text{O}_2$	End-tidal pressure of oxygen
$\dot{Q}$	Cardiac output
$S_a\text{O}_2$	Arterial oxygen saturation
$S_k\text{BF}$	Skin blood flow
$S_p\text{O}_2$	Peripheral oxygen saturation
TLC	Total lung capacity
VC	Vital capacity
$\dot{V}\text{O}_2$	Oxygen consumption
V/Q	Ventilation/perfusion
V_t	Tidal volume

1 Introduction

Since the elevation of atmospheric oxygen levels on Earth, there has been a notable surge in the proliferation of animal species, including humans. With this advancement, eukaryotic organisms must strike a delicate equilibrium between fulfilling their cellular needs and mitigating oxidative damage from oxygen exposure (Hsia et al. 2013). Initially emerging in aquatic environments, life eventually transitioned to terrestrial habitats, partly driven by improved oxygen availability (Long and Gordon 2004). This transition prompted the development of lungs to optimize oxygen utilization (Hsia et al. 2016). As populations of various organisms grew, the competition for food and shelter increased. Some organisms, like penguins and ichthyosaurs, ventured back into the ocean depths (Kelley and Pyenson 2015), while others, such as humans, settled in high-altitude lands (Witt and Huerta-Sánchez 2019) and probably harvested food in the sea (Ilardo and Nielsen 2018). High-altitude and aquatic environments are characterized by reduced oxygen availability, which can result in diminished oxygenation within tissues depending on the duration of exposure. It is termed hypoxia when these oxygen levels are insufficient to maintain adequate homeostasis. Hypoxia poses a significant threat to organisms that depend on aerobic metabolism for energy production, as anaerobic metabolism or

metabolic depression provides limited effectiveness in such conditions (McClelland and Scott 2019).

The capacity to not only survive but thrive in hypoxic conditions suggests a certain level of hypoxia tolerance. In response to the hypoxic aquatic environment, diving birds and mammals have evolved the diving response. This response includes bradycardia and peripheral vasoconstriction (Butler and Jones 1997). These adaptations reduce oxygen consumption ($\dot{V}O_2$), thus facilitating the maintenance of oxygen stores within the lungs, bloodstream, spleen, and muscles (Hochachka 1998; Kelley and Pyenson 2015). Adaptation to hypobaric hypoxia is different in animals and humans as it is mainly due to a complex interaction of sensing mechanisms and signal transduction that are affected by exposure time (Hochachka 1998). Interestingly, recent research indicates a potential shared origin of the defensive mechanisms against hypoxia in humans in both hypoxic environments (Holmström 2022).

Throughout history, humans have engaged in breath-hold diving, also known as freediving, for various purposes, such as gathering food or obtaining goods. It could be speculated that it dates even to the times of archaic humans (Standen et al. 1997; Trinkaus et al. 2019). One example of a population that kept this long-standing tradition of breath-hold diving is the Ama of Korea and Japan. The Ama have honed their diving and underwater exploration skills for over two

thousand years (Minoru 1965). Another example of a population renowned for its seafaring and diving prowess is the Bajau people, who have been actively engaged in breath-hold diving for more than 500 years (Nimmo 1968; Sather 1997). Other examples include divers in ancient Greece (1200 B.C. - 323 B.C.) who undertook diving expeditions to gather shellfish and sponges, contributing to their livelihoods (Ioannidou 2014). Additionally, during times of war, Greek divers played a strategic role by inspecting enemy ship hulls and even sabotaging them (Frost 1968).

1.1 Modern Freediving activities

Freediving is a widely enjoyed recreational activity around the world. It could be broadly categorized into four activities: professional freediving, recreational freediving or snorkeling, spearfishing, and competitive freediving (Fig. 1). However, as participation swells, so does the proportional escalation of associated risks (Pollock 2008). Among these risks, the hypoxic blackout (BO) emerges as particularly dangerous. A hypoxic BO implies the loss of consciousness due to oxygen deprivation and carries, with it, the inherent threat of drowning (Craig 1961). The risk is exceptionally high during solo dives where assistance is impossible.

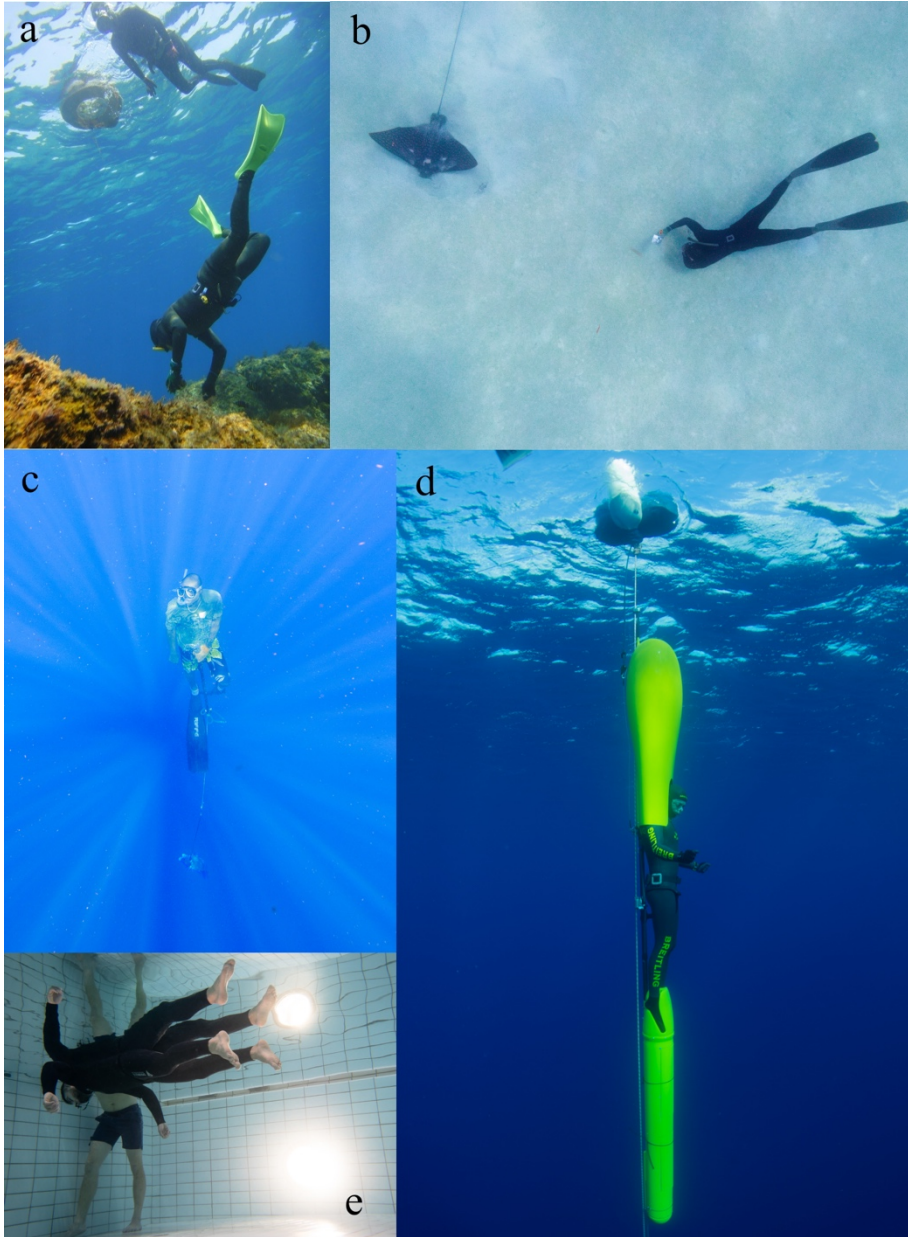


Fig. 1. Professional Ama divers (a) working in pairs for safety in Nanasujima, Japan. Snorkeler (b) taking a photo of a spotted eagle ray and spearfisher (c) with his catch in San Andres Island, Colombia. Freediver in a No-Limits dive in Santorini, Greece. Freediver during static apnea competition (e) in Stockholm, Sweden. Photos by Erika Schagatay (a), the author (b, c), Herbert Nitsch (d), and Nicole Edensbo (e).

Professional freedivers

Professional freediving involves populations extracting resources from the seabed and engaging in spearfishing to secure sustenance and income. Examples of this practice include the Ama divers (Fig. 1a) in Korea and Japan, as well as Indonesia's Bajau community. Moreover, many coastal communities worldwide rely on spearfishing (Fig. 1c) as a vital source of economic support and nutritional sustenance for their families.

Recreational freediving

The population of recreational freedivers is complex since freediving often serves as a recreational pursuit in various tourist destinations. The group encompasses a diverse composition with varying levels of breath-hold training, and many individuals engage in this activity, also called snorkeling (Fig. 1b), even without formal instruction. While it is likely the largest group, determining the exact number of individuals at risk of BO within this population is difficult, given that virtually anyone equipped with essential gear can venture into the aquatic environment without supervision. It is worth noting that snorkeling is associated with a significant portion of drowning-related deaths (77.8%), with hypoxic BO potentially contributing to at least 12.5% of these cases (Lippmann 2019; Dunne et al. 2021).

Spearfishers

Spearfishing has grown popular as a sport, with the primary objective being fish capture (Fig. 1c). According to statistics, spearfishers account for 19.4% of all freediving-related fatalities (Dunne et al. 2021). However, it is difficult to assess how many of these deaths are exclusively due to BO. The increased risk of BO among spearfishers is closely linked to their pursuit of prey. Often, they separate from the main group during their dives, leading to solo freediving scenarios.

Competitive freediving

Freediving has its roots in activities such as spearfishing and collecting goods from the seabed. Still, it has evolved into various competitive sports, such as underwater target shooting, hockey, rugby, and synchronized swimming. While breath-holding skills are a valuable tool in these sports, the goal in competitive freediving is the duration or the distance covered. Competitive freediving involves staying underwater longer, swimming further distances, and diving deeper than ever in a single breath. The duration, distance, and depth that competitive freedivers achieve are unbelievable (Table 1). The deepest discipline is "No-limits," where the diver descends with the help of ballast and returns to the surface with the aid of a floating device (Fig. 1d). The record for the deepest dive in this discipline is 214 meters (Table 1), which was achieved in 2007. However, in 2012,

the same freediver attempted a remarkable depth of 253 meters, but unfortunately, this venture resulted in severe decompression sickness for the athlete (Forni 2015).

Competitive freediving is a rapidly growing sport. Two major agencies arrange competitions, the International Association for the Development of Apnea (AIDA) and the World Federation of Underwater Activities (CMAS), with similar rules and guidelines. The disciplines in freediving are categorized based on the goal (time or distance) and swimming techniques (Table 1).

Table 1. Freediving disciplines and world records endorsed by AIDA

Discipline	Technique	WR
Static apnea (STA; Fig. 1e)	Apnea while floating in a pool.	11'35"
Dynamic apnea with fins (DYN)	Horizontal swimming with monofin	321 m*
Dynamic apnea with bifins (DYNBF)	Horizontal swimming with bifins	292 m*
Dynamic apnea without fins (DNF)	Horizontal swimming without fins	250 m
Constant weight (CWT)	Vertical swimming with monofin	136 m†
Constant weight with bifins (CWTBF)	Vertical swimming with bifins	124 m*
Constant weight without fins (CNF)	Vertical swimming without fins	102 m
Free immersion (FIM)	Pulling down and up from a rope	135 m
Variable weight (VWT) [§]	Down in a weighted sled, up by pulling the rope	156 m
NO-limits (NLT) ^{§#}	Down in a weighted sled, up by a floating device	214 m

WR, world record; * CMAS sanctioned record; † same record for AIDA and CMAS; § no longer considered competition disciplines, # no longer endorsed by any agency.

However, amidst the risks associated with freediving, the human body possesses physiological defense mechanisms that manifest

during underwater submersion. One of them is the diving response. This response enables the body to conserve oxygen by reducing oxygen demands.

1.2 Freediving integrative mechanisms

1.2.1 The Diving response

The diving response in various diving animals, including humans, is a crucial defense mechanism against the inherent dangers of submersion. The diving response involves a series of physiological responses such as heart rate (HR) reduction, vasoconstriction, and systemic hypertension. Among these responses, the HR response is the one that has been most extensively studied in breath-holding research, as it reflects the overall response and is easy to measure.

1.2.1.1 Diving bradycardia

The beginning of a breath-hold is characterized by a short period of increased HR, i.e., tachycardia, followed by a reduction in HR until a plateau is observed (Schuitema and Holm 1988; Fig. 2). The decrease in HR, i.e., bradycardia, is boosted by whole-body immersion (Harding et al. 1965; Campbell et al. 1969; Moore et al. 1972; de Bruijn et al. 2009; Costalat et al. 2013) as well as facial immersion in cool water (Kawakami et al. 1967; Whayne and Killip 1967; Ferrigno et al. 1997). Diving-induced bradycardia persists during apnea and swimming (Craig and Medd 1968a; Asmussen and Kristiansson 1968) and static cycling (Craig and Medd 1968a; Asmussen and

Kristiansson 1968; Bergman et al. 1972; Paulev and Hansen 1972; Andersson et al. 2004). Cold water immersion further enhances the magnitude of diving-induced bradycardia (Paulev et al. 1990; Schagatay and Holm 1996).

Breath-hold training also affects bradycardia, leading to stronger bradycardic responses in trained freedivers (Schagatay et al. 2000; Delapille et al. 2002; Lemaître et al. 2013). Bradycardia during apnea is regulated by various neural mechanisms, including input from facial cold receptors, atrial receptors, baroreceptors, pulmonary stretch receptors, and carotid chemoreceptors (Foster and Sheel 2005). Additionally, venous return to the heart or preload may be positively influenced by negative intra-thoracic pressure, as body immersion plays a role in this response (Song et al. 1969). The inputs from the different receptors, during breath-holding, increase the activity in several central nervous system centers, specifically, fibers from the cardiovascular inhibitory center (Bove et al. 1973; Andersson et al. 2002, 2004; Lemaître et al. 2013), which increases the parasympathetic activity on the sinoatrial node, resulting in a pronounced drop in the HR, approximately a reduction from 20-40%.

Diving bradycardia exhibits distinct phases (Fig. 2). The initial phase is marked by a rise in HR, typically lasting for 25 to 30 seconds, followed by a sustained bradycardic state, and sometimes could exhibit a third phase (Perini et al. 2010). The third phase, which is

characterized by a slight decrease in heart rate, has been associated with physiological stress (Sivieri et al. 2015) and blood oxygen desaturation (Costalat et al. 2015). The third phase could be hypoxia-mediated by carotid bodies by increasing vagal activity (Lemaître et al. 2008).

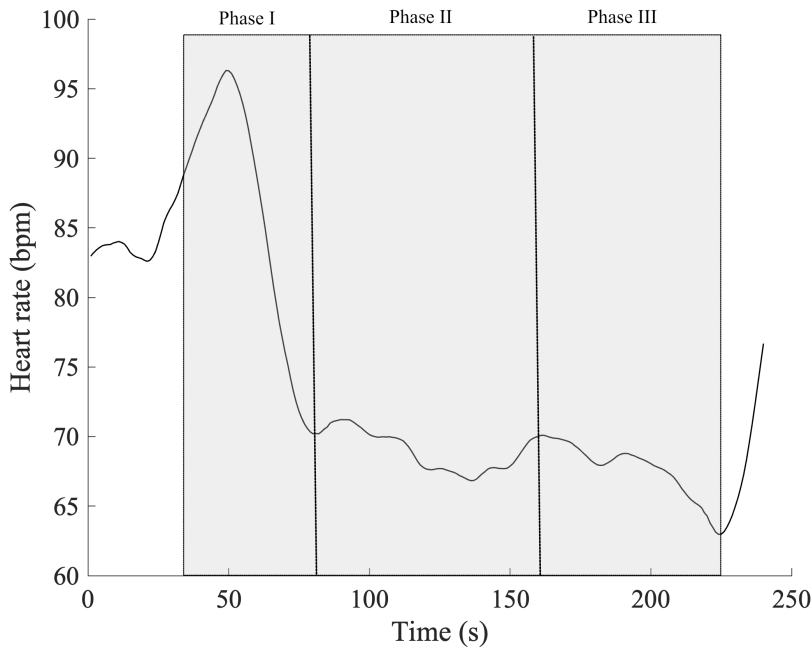


Fig. 2. Time course of breath-hold diving bradycardia. The grey box indicates breath-hold time.

Diving-induced bradycardia, as demonstrated in several studies (Wolf 1964; Schagatay and Andersson 1998; Ferretti 2001; Andersson et al. 2002; Andersson and Evaggelidis 2009; Costalat et al. 2015), has an oxygen-conserving effect. Alongside the bradycardia, there is a decrease in cardiac output. Some studies have reported a reduction in

stroke volume (Kawakami et al. 1967; Gross et al. 1976; Baković et al. 2006; Marabotti et al. 2008, 2009, 2013; Sivieri et al. 2015; Hoiland et al. 2017; Magnani et al. 2018; Eichhorn et al. 2018; Taboni et al. 2018) while others have found no change (Paulev et al. 1990; Ferrigno et al. 1997; Costalat et al. 2017), or even an increase (Ghiani et al. 2016). These differences may be partially due to variations in the number of participants, dietary status, and variations in measurement methods. Still, most studies agree that the primary effect of diving bradycardia combined with stroke volume reduction is decreased cardiac output.

1.2.1.2 Peripheral vasoconstriction

The second component of the diving response is the peripheral vasoconstriction, which is mediated by sympathetic activation (Olsen et al. 1962; Wolf 1964; Harding et al. 1965; Kawakami et al. 1967; Heistad et al. 1968; Campbell et al. 1969; Schagatay and Andersson 1998; Foster and Sheel 2005). The primary effect of vasoconstriction is to redistribute the blood flow to critical organs (Lin 1982) like the brain (Willie et al. 2015; Bruce et al. 2016; Bain et al. 2017; Keil et al. 2018; Vestergaard and Larsson 2019; Bouten et al. 2020; AlSalahi et al. 2020; McKnight et al. 2021), lungs (Prediletto et al. 2009; Fahlman et al. 2009) and heart (Pingitore et al. 2008; Eichhorn et al. 2018). Given that the diving response necessitates the simultaneous engagement of sympathetic and parasympathetic components, it has prompted the

formulation of the "autonomic conflict" theory as a cause of cardiac arrhythmias during breath-hold diving (Panneton and Gan 2020).

In summary, a strong diving response leads to an oxygen-conserving effect, as a smaller portion of the body uses the blood oxygen supply (Andersson and Schagatay 1998; de Bruijn et al. 2009). Additional factors contribute to the complex mechanisms of cell hypoxia within breath-hold diving.

1.2.2 The splenic contraction

In addition to the diving response, there is another protective mechanism against hypoxia. During breath-hold dives, the spleen contracts and releases erythrocytes into the circulation, which enhances the circulating oxygen storage and thereby prolongs the apnea (Schagatay et al. 2001; Espersen et al. 2002; Elia et al. 2021a). The spleen has myosin proteins in the walls of arteries and veins and even in the splenic sinuses (Pinkus et al. 1986), meaning the spleen can regulate its volume from within (Stewart and McKenzie 2002). The mechanism of the spleen contraction in humans has been linked to sympathetic activation at the very beginning of apnea (Palada et al. 2007), with a further contraction that seems related to the severity of hypoxia (Richardson et al. 2009). However, it was observed that eupneic hypoxia incited a milder degree of contraction when contrasted with apneic hypoxia (Lodin-Sundström and Schagatay

2010). It was later found that hypercapnia (Richardson et al. 2012) contributes slightly to the Hb increase even in the absence of hypoxia.

To summarize, the initial contraction of the spleen can be attributed to sympathetic activity triggered by breath-holding. Hypoxia, on the other hand, could lead to an additional contraction of the spleen, even in the absence of apnea, while hypercapnia has a comparatively lesser impact on this contraction.

1.2.3 Control of breathing

During normal breathing, the respiratory rhythm is regulated by involuntary chemosensory, mechanical, and voluntary behavioral input (Caruana-Montaldo et al. 2000). A central controller processes these inputs, including the pons, medulla, and other brain regions. Subsequently, an output signal is delivered to the respiratory muscles (Skow et al. 2015). The brain cortex can override, to some extent, the function of the brain stem, as happens during voluntary hyperventilation, and other parts of the brain, such as the hypothalamus and the limbic system, can also affect the respiratory rhythm (Homma and Masaoka 2008).

The chemosensory input is mediated by feedback reflexes from central and peripheral chemoreceptors that respond to changes in the arterial pressure of carbon dioxide ($P_a\text{CO}_2$), pH, and hypoxemia (low oxygen levels in the blood). The most important central

chemoreceptors are near the medulla's ventral surface. They are surrounded by brain extracellular fluid and respond to increases in the hydrogen ion (H^+) concentration, which stimulates ventilation. As the blood-brain barrier is relatively impermeable to H^+ and bicarbonate (HCO_3^-), the main controller is the P_aCO_2 , as CO_2 diffuses freely to the brain extracellular fluid and into the cerebrospinal fluid, liberating H^+ and stimulating ventilation (West and Luks 2021).

Central chemoreceptors are insensitive to reduced P_aO_2 levels (Nattie and Li 2012). However, peripheral chemoreceptors sense changes in blood oxygenation.

Peripheral chemoreceptors are in the carotid and aortic bodies. The carotid bodies are of greater importance, as they comprise glomus cells (type I) and sheath cells (type II). Type I cells are close to the end of the afferent carotid sinus nerve. Peripheral chemoreceptors respond to decreases in P_aO_2 and pH and increases in P_aCO_2 . The firing rate of the carotid bodies depends mainly on the levels of P_aO_2 , and it exhibits a non-linear relationship, as the firing rate is very slow with changes in the range of 100-500 mmHg. The firing rate increases substantially when P_aO_2 drops below 100 mmHg. Peripheral chemoreceptors are less sensitive than central chemoreceptors to changes in P_aCO_2 , and only the carotid bodies respond to changes in blood pH (West and Luks 2021).

Mechanical receptors are also involved in controlling breathing. Pulmonary stretch receptors hold particular significance. They are located within the tracheobronchial tree and lie within the airway smooth muscle. Their primary role is to terminate inspiration when the lungs are inflated (Hering-Breuer inflation reflex). So, their activity is reduced with low lung volumes (West and Luks 2021).

When breathing stops, the $P_a\text{CO}_2$ increases and stimulates the respiratory drive (Forster et al. 2012). In addition to the CO_2 stimulus, when $P_a\text{O}_2$ drops below 50 mmHg, it will induce maximal activity in carotid and aortic bodies chemoreceptors, further increasing the ventilatory drive (Parkes 2006). The breath-hold diver interprets this escalated neural activity as an intense "urge" to breathe, often accompanied by contractions of the diaphragm and other respiratory muscles (Hentsch and Ulmer 1984; Agostoni 1963). These respiratory muscle contractions are called involuntary breathing movements (IBM) or "contractions" within the freediving community. The start of these IBM indicates the physiological breaking point during apnea (Fowler 1954; Agostoni 1963; Hill 1973). With increased respiratory center stimulation, this physiological alert prompts the need to resume breathing. Nonetheless, trained freedivers can prolong this phase (termed the "struggle phase") through voluntary control. How they can hold their breath way beyond the physiological breaking point is speculative. It could be related to a balance between effort

(apnea duration) versus return (discomfort) integrated with the central respiratory system (Parkes 2006).

1.3 Hypoxia

Breath-holding implies breathing interruption. Therefore, breath-holding gradually depletes the body's oxygen reserves, ultimately resulting in cell hypoxia. Additionally, hypoxia is directly influenced by hypoxemia and the blood flow to specific tissues (Rodríguez-Roisin and Roca 2005). The human brain is exceptionally vulnerable to hypoxia, as it lacks a substantial local oxygen reserve, making it heavily reliant on a continuous supply of oxygen-rich blood (Oddo et al. 2014). The onset and progression of brain hypoxia is influenced by other factors besides hypoxemia, like $\dot{V}O_2$, hemoglobin concentration (Hb), and cardiac output ($\dot{Q}$). Essentially, hypoxia severity arises from the interaction between oxygen delivery ($\dot{D}O_2$) and $\dot{V}O_2$ (Bloos and Reinhart 2005). The specific interplay of these factors determines the extent and rate at which hypoxia manifests.

$\dot{D}O_2$ is the product of $\dot{Q}$ and arterial oxygen content (C_aO_2). Hence, the complete expression is as follows:

$$\dot{D}O_2 = \dot{Q} \cdot \{(SaO_2 \cdot [Hb] \cdot 1.39) + (0.003 \cdot PaO_2)\} \quad (\text{Eq. 1})$$

Here, S_{aO_2} represents the fraction of oxygen saturation. The contribution of P_{aO_2} is negligible at sea level, but its importance increases during dives where hydrostatic pressure rises.

1.3.1 Hypoxemia

The lungs serve as a vital oxygen reservoir in humans (Overgaard et al. 2006; Ferretti et al. 2012; Schagatay et al. 2012). The breath-hold duration gradually reduces the alveolar partial pressure of oxygen (P_{AO_2}), decreasing the driving pressure into the capillary (Craig and Harley 1968). Research findings have demonstrated a positive correlation between the initial lung volume and the P_{AO_2} at the end of apnea (Hong et al. 1971) and the rate of blood oxygen desaturation (Findley et al. 1983). While the lungs serve as vital oxygen stores, it is essential to note that the blood and tissues also play a crucial role in oxygen storage (Cherniack and Longobardo 1970; Liner and Linnarsson 1994; Elia et al. 2019). As oxygen stores are finite during apnea, maintaining normoxemia depends mainly on the size of this reserve. Total oxygen stores are around 1120 to 2000 ml (Table 2) in a 70 kg person (Rahn 1964; Cross et al. 1968; Cherniack and Longobardo 1970; Lumb 2021).

Typically, elite freedivers engage in a technique referred to as "lung-packing" (glossopharyngeal insufflation), which entails increasing vital capacity (VC) and total lung capacity (TLC) after a maximal inhalation (Örnhammar et al. 1998). The increase in TLC can reach 11 L

in some freedivers (Lindholm and Nyren 2005; Overgaard et al. 2006; Seccombe et al. 2006), theoretically contributing to the oxygen storage capacity of ~**1600** ml. Furthermore, some elite freedivers have 27% more myoglobin concentration in type I fibers than healthy controls (Elia et al. 2019), contributing to ~**305** ml. Several studies did not find a difference in Hb between controls and freedivers (Richardson et al. 2005; Baković et al. 2005; Fernández et al. 2017). Recently, it was found that elite freedivers have 5 g/L more Hb than matched controls (Elia et al. 2019), and that could add ~**35** mL of oxygen.

Similarly, investigations have shown no differences in blood volume (Prommer et al. 2007). Consequently, the current evidence does not support the notion that apneic training results in larger blood oxygen stores. Splenic contraction could add 4 g/L of Hb to the circulation, contributing to ~**28** ml (Richardson et al. 2005). Accordingly, the theoretical oxygen stores of an elite freediver could be ~**3000** ml.

Table 2. Oxygen stores in a 70-kg man at sea level

	Estimate (mL)				Form of storage
	Rahn 1964	Cross 1968	Cherniack 1970	Lumb 2021	
Lungs	370*		500*	450*	Dissolved gas
Blood	880		1200	850	Dissolved gas & oxyhemoglobin
Tissues	56		300†	50	Dissolved gas
Mb	240			200	
Total	1546	1120	2000	1550	

Mb, bound to myoglobin * Measured at functional residual capacity (FRC). † Includes bound to myoglobin.

The duration of apnea is not only influenced by oxygen reserves but is also subject to the $\dot{V}O_2$ during freediving, which is impacted by several factors. In the context of static apnea, where no physical exertion is involved, $\dot{V}O_2$ varies between 220 – 264 ml/min at sea level (Otis et al. 1984). Adding the swimming effort to the apnea widens this range to 350 to 800 ml/min (Craig and Medd 1968b; Ferretti et al. 1991). The increase in $\dot{V}O_2$ can be attributed to additional skeletal muscle work (Andersson et al. 2002, 2004). Furthermore, prolonged

exercise before apnea can amplify oxygen desaturation due to excess post-exercise $\dot{V}O_2$ (Lindholm and Gennser 2005).

Water has a significantly higher heat absorption capacity than air, and its thermal conductivity is substantially greater. Upon cold water exposure, the body increases heat production by shivering, sympathetic excitation of heat production (Castellani and Young 2016), and thyroxine secretion (Maushart et al. 2021). The changes induced by cold water exposure are known as thermal stress. Thermal stress contributes to an overall increase in $\dot{V}O_2$. Without insulation, freediving in 10°C water temperatures could elevate $\dot{V}O_2$ by as much as 43% when contrasted with freediving in water maintained at 22-26°C for at least 25 minutes (Kang et al. 1965).

$\dot{V}O_2$ also increases as a function of diving depth, although not linearly. As the depth increases, the buoyancy is reduced, causing greater exertion in the initial meters closer to the surface. Therefore, it follows that $\dot{V}O_2$ is relatively elevated during shallower dives characterized by greater buoyancy, in contrast to deeper dives with lower buoyancy (Craig and Medd 1968b). When returning to the surface, the $\dot{V}O_2$ should be relatively increased in the deeper part of the dive, compared with the shallower part, where greater buoyancy is synergistic with the swimming effort.

Elite freedivers seem to have a lower capacity for mitochondrial $\dot{V}O_2$ than non-freedivers (Kjeld et al. 2018). Some techniques like yoga and

meditation (frequently used before freediving) can reduce resting $\dot{V}O_2$ (Sarang and Telles 2006). Moreover, fasting can reduce the metabolic rate even below the resting metabolic rate (Schagatay and Lodin-Sundström 2014).

The resting $\dot{V}O_2$ of an elite freediver weighing 77 kg has been estimated at around **230 ml/min** (Lundberg et al. 2023). In a theoretical scenario, considering the calculated oxygen reserves, this exceptional freediver could endure a breath-hold lasting over **13** minutes, granted he withstands the urge to breathe, before fully depleting his oxygen stores (Herber 1948; Stockard et al. 2007). In this theoretical model of hypoxemia, the concept of tolerance to asphyxia is omitted, as is the consideration of consciousness. It is anticipated that profound cerebral hypoxia will lead to a loss of consciousness before consuming all oxygen stores.

1.3.1.1 Gas Exchange and depth

Although not evaluated in this thesis, depth also induces hypoxemia by different mechanisms, and it is worth reviewing basic concepts.

P_{AO_2} in humans can be expressed as:

$$P_{AO_2} = F_i O_2 (P_B - P_{H_2O}) - \frac{P_A C O_2}{\left(\frac{V C O_2}{V O_2} \right)}$$

(Eq. 2)

Where F_{iO_2} is the inspired fraction of oxygen (0.21 in ambient air), P_B is the barometric pressure in mmHg, P_{H_2O} is the partial pressure of water vapor, P_{ACO_2} is the alveolar partial pressure of CO_2 , V_{CO_2} is the volume of CO_2 released, and V_{O_2} is the volume of air absorbed during respiration. At sea level, oxygen diffuses into the pulmonary capillary, resulting in a $P_{aO_2} \sim 100$ mmHg (West and Luks 2021).

During freediving, the hydrostatic pressure increases, raising P_{AO_2} and P_{aO_2} levels (Table 3; Eq. 2). However, in a study of six divers, two had lower P_{aO_2} levels at a depth of 42 meters than the other four (Bosco et al. 2018). These two participants recorded P_{aO_2} values of 76 mmHg and 61 mmHg, respectively, alongside a S_aO_2 of 94%. In a case study involving a single participant (Scott et al. 2021), the second dive at the same depth (60 meters) showed a P_{aO_2} of 249.8 mmHg, 22% less than the value of the first dive. This data highlights that a hyperbaric environment does not invariably lead to higher P_{aO_2} levels, as variations are evident among individuals and within the same individual across different dives.

Table 3. Summary of arterial blood gases at depth in freediving.

Reference	n	D	T	P _a O ₂	S _a O ₂	P _a CO ₂
Qvist et al. 1993	5	5	24	141 ± 24	99.0 ± 0.7	46.6 ± 2.4
Muth et al. 2003* †	2	20	60	225.3 ± 0.5	N.R.	41.2 ± 2.1
Bosco et al. 2018	6	42	45	197.8 ± 103.7	98.0 ± 3.0	42.7 ± 12.3
Scott et al. 2021 †	1	60	65	321	N.R.	47.3
Barković et al. 2023	3	60	74	209.5 ± 16.2	100	54.1 ± 1.6

n, Number of participants; D, depth in meters; T, average time in seconds of apnea duration when the sample was taken; P_aO₂, arterial pressure of oxygen in mmHg; S_aO₂, arterial saturation of oxygen in percentage; P_aCO₂, arterial pressure of carbon dioxide in mmHg. * Participants performed lung packing. † Participants performed hyperventilation.

Hydrostatic pressure also compresses the lungs and the thoracic cage (Ferretti 2001). Different elastance may cause varying volume changes in the chest wall and the lungs during a dive (West and Luks 2021). Lung volume reduction is larger than chest wall volume reduction. Additionally, individual geometrical differences in lung shape (Torres-Tamayo et al. 2018) could lead to heterogeneous reductions in intra-pleural pressure (Pasticci et al. 2020). The overall effect is an increase in the transpulmonary pressure with an increase in the hydrostatic gradient between pulmonary capillaries and pulmonary interstitial tissue. This pressure difference increases the net flow of liquid from the capillary to the interstitial tissue (Starling 1896).

Additionally, during immersion, as depth increases, there is blood redistribution to the thorax (Lindholm and Lundgren 2009), which

increases the intrathoracic blood volume (ITBV). Freedivers have a higher incidence of patent foramen ovale, which could be due to increased pulmonary pressure (Kelly et al. 2022). Increased ITBV and pulmonary hypertension could induce recruiting intrapulmonary arteriovenous shunts (Lovering et al. 2006) and increase the amount of right-to-left shunt (Scherhag et al. 2005). Furthermore, it's worth noting that as the alveoli shrink in response to increased hydrostatic pressure during breath-hold diving, impacting ventilation/perfusion (V/Q) relationship. V/Q inequities can contribute to hypoxemia (Rodríguez-Roisin and Roca 2005). Therefore, hypoxemia can be induced by depth through additional mechanisms.

In summary, the final P_{aO_2} level is influenced by many factors and has the potential to decline significantly. For instance, the lowest recorded value after a breath-hold dive is 18.0 mmHg (Bosco et al. 2020).

1.3.2 Reduced perfusion

During breath-hold diving, the activation of the diving response leads to a notable decrease in cardiac output. Consequently, there is a significant reduction in the blood volume that the heart pumps into the circulatory system during the apnea period. However, thanks to strong vasoconstriction in peripheral regions, perfusion is prioritized for hypoxia-sensitive organs, such as the brain and the heart (Ferretti 2001). During freediving, in particular, multiple studies have

demonstrated the prioritization of brain perfusion (Kjeld et al. 2009; Joulia et al. 2009; Ratmanova et al. 2016; Vestergaard and Larsson 2019). It is worth noting that hyperventilation, a breathing pattern commonly employed before apnea diving, has been shown to reduce brain perfusion in clinical settings (Carrera et al. 2010; Bagwell et al. 2016). Nevertheless, the specific impact of hyperventilation on brain perfusion before breath-hold diving remains unclear.

In general, brain perfusion is prioritized during freediving; the peripheral perfusion and muscle flow are reduced during breath-holding, leading to muscle hypoxia, an effect that is enhanced by apnea training (Palada et al. 2007; Bouten et al. 2020). These findings suggest that freedivers rely on myoglobin for muscle metabolism (Elia et al. 2019).

Regardless of all these physiological defense mechanisms and adaptations to hypoxia, freediving poses inherent risks, such as hypoxic BO, a condition characterized by a loss of consciousness due to inadequate oxygen supply to the brain.

1.4 Breath-hold diving blackout.

The occurrence of loss of consciousness during breath-hold diving is referred to by various terms, including shallow water BO, freediving BO, hypoxic syncope, or simply hypoxic BO. Among freedivers, the loss of consciousness is commonly known as a BO. A classic paper

describing BO showed that breath-hold divers could lose consciousness underwater without any warning (Craig 1961). Factors such as hyperventilation and physical exertion were identified as risks for BO (Craig 1961). The physiopathology of the BO was ascribed to low levels of $P_a\text{CO}_2$, which were unable to trigger the respiratory stimulus, causing prolongation of the apnea duration and exposing the brain to profound hypoxia. Consequently, in this thesis, the term hypoxic BO is used to identify its occurrence related to breath-hold diving (Craig 1961; Boyd et al. 2015).

1.4.1 Description of the problem

Individuals interested in breath-hold diving often have easy access to aquatic environments like swimming pools, lakes, and the open ocean, where strict regulation or supervision is frequently lacking. The combination of unrestricted access and lack of proper supervision exposes a considerable portion of the population to the potential dangers of breath-hold-related BO. Furthermore, the absence of adequate oversight means that fatalities, particularly among those engaging in solo diving, can often go unnoticed. The actual number of deaths related to hypoxic BO while breath-hold diving is unknown. The Divers Alert Network (DAN) reported 726 deaths related to breath-hold diving from 2004 to 2017. This number is likely an underrepresentation of the total number of incidents (DeWitt et al. 2019). Australia has a long-term registry of breath-hold-

related fatalities and considers that hypoxic BO is the cause of at least 12% of the fatalities (Lippmann 2019). A recent systematic review on snorkeling-related drowning (Dunne et al. 2021) revealed that men are overrepresented in fatal accidents (83%). The age groups most represented in fatal drownings are individuals aged 19-34 years (33%) and those over 65 years old (27%). The most frequent activity was recreational snorkeling activities, with 78% of the drowning fatalities, followed by spearfishing-related incidents (19%). An Australian report described that 65% of the victims were found to be overweight or obese (Lippmann 2019). Numerous reports emphasize the fact that a significant portion of fatalities (ranging from 50% to 100%) occurred when the individuals were alone at the time of the incident (Edmonds and Walker 1999; Walker et al. 2009; Lippmann et al. 2011, 2013; Lippmann and Lawrence 2012; Denoble and Moore 2018; DeWitt et al. 2019; Lippmann 2019; Dunne et al. 2021). In competitive freediving, the incidence of fatalities is notably reduced due to stringent safety protocols and regulations (Denoble and Moore 2018). Promoting responsible practices, most freediving organizations advocate for comprehensive training and the implementation of the buddy system (ensuring a partner is near to offering aid). Furthermore, competitive freedivers must obtain medical clearance before engaging in competitions.

1.4.2 Risk factors

Traditionally, hyperventilation has been considered a leading cause of hypoxic BO as it reduces the $P_a\text{CO}_2$ levels and delays the urge to breathe. The delayed urge to breathe could then increase the duration of the breath hold dive, so the diver might develop profound hypoxemia without noticing (Craig 1961, 1976; Edmonds and Walker 1999; Lippmann and Pearn 2012). Additionally, hyperventilation reduces brain blood flow (Viola et al. 2012; Immink et al. 2014), which could also be a factor for brain hypoxia. Moreover, another risk factor could be prolonged exercise, leading to an elevated lipid metabolism with a decreased respiratory exchange ratio (Lindholm and Gennser 2005). Regarding depth, slower CNF dives appear to carry a higher risk of BO. In contrast, during CWT dives, a faster active descent followed by a prolonged passive descent phase has been associated with an elevated risk (Poiret et al. 2020). Some case reports present alternative explanations that diverge from hypoxia. These reports propose that cardiovascular collapse resulting from extreme bradycardia and vasodilation (Joulia et al. 2013; Bouten et al. 2020), along with arrhythmias (Wolf 1964; Joulia et al. 2013), may serve as contributing factors to BO.

In contrast, some species of marine diving mammals, like bottlenose dolphins, Harbour porpoises, and Weddell seals, appear to remain unaffected by BO or any other neural impairments while executing dives that far surpass the capabilities of average human divers

(Williams et al. 2008; Ponganis 2011). This exceptional ability could be attributed to the increased expression of neuroglobin, an oxygen-binding protein in the brains of marine mammals. Neuroglobin functions as a second line of defense during hypoxic conditions (Williams et al. 2008), allowing them to reach extreme hypoxemia without experiencing BO (Ponganis 2011).

Despite identifying certain risk factors associated with BO incidents during breath-hold diving, our comprehension of its root cause still needs to be completed, and our knowledge of effective prevention measures needs to be improved. Essential queries that arise include:

To what extent does the diving pattern of recreational divers influence the associated risk? In other words, does engaging in consecutive dives introduce a different level of risk compared to single dives?

Does individual variability impact susceptibility to BO occurrences?
Can the diving response and spleen contraction protect against BO?
How does hyperventilation affect oxygen stores, diving response, and spleen contraction?

2 Aims

This thesis aims to improve understanding of the physiological mechanisms underlying the development of hypoxia in breath-hold diving, particularly its connection with hypoxic blackout. This research involves exploring contributing risk factors and the efficacy of protective responses in mitigating these risks. It also aims to develop strategies that can effectively identify individuals who may be at an elevated risk of faster oxygen desaturation and, in turn, formulate strategies to prevent the occurrence of BO. Developing a test to identify individuals at risk of severe hypoxemia was one strategy to explore.

To accomplish this overarching objective, the research has been divided into the following specific aims:

- a) To enhance our understanding of the risk factors precipitating BO events during breath-hold diving.
- b) To explore the adaptive dynamics of protective responses to hypoxia across diverse physiological conditions.
- c) To study the serial diving pattern typical of recreational freedivers, with a particular focus on the effect of hyperventilation in oxygen desaturation.
- d) To investigate methods for detecting individuals at a higher risk of oxygen desaturation and to elucidate potential

strategies for preventing hypoxic BO in real freediving scenarios.

The specific aims were answered in four different studies:

- Study I Is splenic contraction induced solely by hypoxia fast enough to protect freedivers during extended breath-holding periods?
- Study II How does hyperventilation affect oxygenation, the diving response, and spleen contraction within the context of simulated serial freediving?
- Study III To explore the relationship between force sensor data attached to a chest belt and pre-apnea End-tidal CO₂ pressure ($P_{ET}CO_2$) values in freedivers.
- Study IV Can a static apnea ramp test identify freedivers at risk of faster blood oxygen desaturation?

3 Methods

3.1 Participants

In the present thesis, a cohort of 147 healthy volunteers, 57 women and 90 men, participated (Table 4). All participants were distinct individuals, except for two male freedivers who participated in **Studies III** and **IV**.

Most participants were active freedivers at different levels of experience and training (Table 4). To categorize participants and control for the impact of previous apnea training, a classification system (Schagatay 1996) was employed to group participants into five distinct categories:

- I. No prior experience in breath-hold diving.
- II. Previous involvement in breath-hold diving, although not within the past year.
- III. Regular breath-hold training for up to 2 hours per week during the past year.
- IV. Regular breath-hold training of more than 2 hours per week, but less than 2 hours per day, during the past year.
- V. Daily breath-hold training of at least 2 hours per day during the past year.

The participants in **Study I** and **II** maintained an active and healthy lifestyle but were not engaged in freediving training. Table 4 presents the characteristics of the participants in the different studies.

Table 4. Characteristics of participants in the thesis (Study I – IV).

Study	<i>n</i>	Population	Sex (f, m)	Age (yr)	Height (cm)	Body mass	Vital Capacity	Cat.*
I	15	Non- freedivers	8, 7	25 ± 3	178 ± 11	77 ± 14	5.2 ± 1.2	I-II
II	18	Non- freedivers	6, 12	30 ± 7	177 ± 8	76 ± 11	5.4 ± 1.1	I-II
III	21	Freedivers	5, 16	44 ± 7	178 ± 9	73 ± 10	5.8 ± 1.2	IV-V
IV	93	Freedivers	38, 55	37 ± 9	175 ± 10	70 ± 13	5.9 ± 1.4	IV-V

Data presented as mean ± SD; f, females; m, males; yr, years; cm, centimeters; kg, kilograms; L, liters. * predominant category (> 80% of the participants).

3.1.1 Ethical considerations

The participants were recruited via convenience sampling. They were provided with written and verbal information about the study's objectives, the right to interrupt participation without consequences, and the protocols governing handling personal information. In all studies, participants formalized their engagement by signing written consent documents. All studies were performed following the declaration of Helsinki, except for previous registration. The Swedish law states that an ethical review must be done if 1) the study protocol involves physical intervention, 2) it is carried out with a method that aims to affect the research participant physically or mentally or involves an obvious risk of harm to them in body or mind, or 3) involves the processing of sensitive personal data. The studies of this thesis needed review mainly on the basis that they involved physical interventions and thereby affected the participants. Safety measures

were taken to mitigate risk as described in the annexed papers. **Study I** was approved by the Regional Ethical Review Board (EPN) of Umeå University, Umeå, Sweden (#2011-453-31M), and **Studies II, III, and IV** were approved by Swedish Research Ethics Authorities (EPM) (#2019-05147).

3.2 Overview of studies

This thesis is based on laboratory and field laboratory studies. Laboratory studies provide a controlled environment for isolating and studying variables one at a time, leading to more accurate data than field studies. However, they may lack real-world applicability, as participants' behaviors in controlled settings can differ from those in natural environments. On the other hand, field studies are typically more generalizable to the actual activity, in this case, freediving. Yet, the limited control over numerous variables can pose challenges in drawing precise conclusions. Using a field laboratory emerges as an approach to achieve more accurate results and ecologically valid studies for diving safety.

Studies I and II were conducted as laboratory studies because the precision required for the intervention and the necessity of measuring a large number of variables made it challenging to attain in a field environment. **Studies III and IV** used a mixed setting combining laboratory and field testing by deploying a mobile laboratory designed for use in proximity to designated freediving training or

competing locations. **Study III** was conducted during a freediving competition for a more ecologically valid setting. The main characteristics of the different studies are presented in Table 5.

Table 5. Characteristics of the studies in the thesis.

Study	Setting	Protocol	Aim
<i>Study I</i>	Laboratory	Breathing 10% Oxygen	To assess spleen contraction speed, its impact on Hb levels, and recovery in response to sudden extreme hypoxia.
<i>Study II</i>	Laboratory	Apnea series with facial immersion	To elucidate the effects of hyperventilation compared to normal ventilation on oxygenation, diving response and spleen contraction during serial simulated dives.
<i>Study III</i>	Field Laboratory	Dry static apnea ramp test	To identify pre-apnea hyperventilation by using a force sensor.
<i>Study IV</i>	Field Laboratory	Static apnea ramp test with facial immersion	To identify individual variables affecting the rate of oxygen desaturation.

3.2.1 Laboratory (Studies I and II)

In **Study I**, participants were exposed to normobaric hypoxia (breathing 10% oxygen) for 10 minutes during a sitting rest, followed by 10 minutes of breathing 100% oxygen. HR, S_pO_2 , and mean arterial blood pressure (MAP) were measured continuously during the protocol. Additionally, the spleen's 3-axial dimensions were assessed using ultrasonic imaging every minute for volume calculations, and venous blood samples were collected before, during, and after the exposure to measure hemoglobin levels.

In **Study II**, participants executed two sets of 5 apneas, each separated by 2-minute intervals, with cold water ($15 \pm 1^\circ\text{C}$) facial immersion to their maximum duration on volition. The control series of apneas were initiated following the instruction to use normal breathing, while the experimental series were preceded by 15 seconds of hyperventilation. The participants were given instructions prior to the test on how to perform hyperventilation based on a previous protocol (de Bruijn et al. 2008). The order of the conditions was weighted between participants, and a total pause of 20 minutes separated the apneas in the two series. Before and after each apnea, end-tidal oxygen and end-tidal carbon dioxide were measured. Continuous monitoring included S_pO_2 , HR, breathing movements, and skin blood flow from 5 minutes before starting the protocol to 5 minutes after. Spleen diameters in two dimensions were also

measured at 15-second intervals across the protocol. The mean room temperature was $23 \pm 1^{\circ}\text{C}$, and the mean barometric pressure was $762 \pm 6 \text{ mmHg}$.

3.2.2 Field laboratory (Studies III and IV)

In **Study III**, the participants completed a static apnea ramp test. The test involved three apneas of increasing durations (1, 2, and 3 minutes) with 2-minute rest intervals between apneas. The participants performed the test without facial immersion and in a sitting position. Slow VC was measured in triplicate. P_{ETCO_2} was measured before and after each apnea. S_{pO_2} and HR were measured continuously 3 minutes before the first apnea until 5 minutes after the last apnea. Chest movements were recorded via a chest belt using a force sensor (Sieber et al. 2022) specially developed to record the frequency and amplitude of chest movements (Fig. 3). Before implementing this methodology, a preliminary pilot study was conducted (Paper III Supplementary Information). The main aim of this pilot study was to evaluate the measurement error of the force sensor when estimating VC by analyzing the data obtained from its signal. In **Study III**, a linear regression was employed to predict the P_{ETCO_2} using the product of amplitude and respiratory rate recorded 60 seconds prior to apnea. The room temperature was $26.9 \pm 2.0^{\circ}\text{C}$.

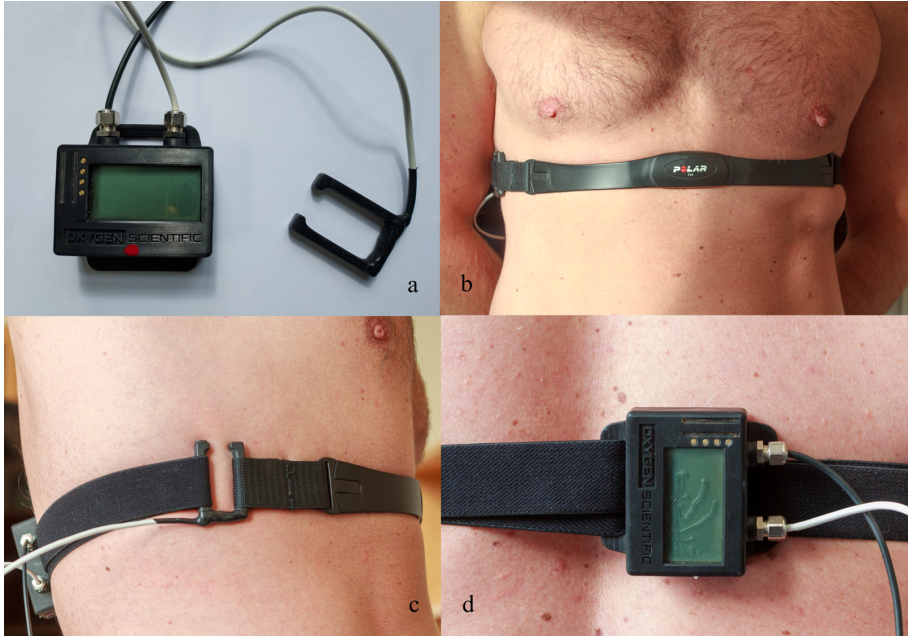


Fig. 3. A custom-made buckle equipped with four strain gauges aligned on the sensor to measure the strain created by the applied force, along with the datalogger (a). Frontal (b) and lateral (c) views of the buckle in its operational position attached to the chest strap. Detail of the datalogger (Sieber et al. 2022) in operational position in the back of the participant (d).

In **Study IV**, the participants performed a static apnea ramp test, which consisted of three apneas of increasing durations (1, 2, and 3 minutes) with 2-minute rest intervals between apneas. The participants performed the apneas with their faces immersed in cold water (14.8 ± 0.8 °C) to simulate diving conditions (Schagatay and Holm 1996). Slow VC was measured in triplicate in a standing position, and spleen dimensions were measured in the sitting position after at least six minutes of rest. VC and spleen dimensions were measured only before the test. $P_{ET}CO_2$ was measured before and

after each apnea. Hb and lactate were measured before the test started and after the last apnea. S_pO_2 and HR were measured continuously 3 minutes before the first apnea until 5 minutes after the last apnea. The room temperature was maintained at 22.9 ± 1.3 °C using a central air conditioning unit. The ambient and water temperatures were controlled before and during every test to maintain a temperature gradient of 10 ± 2 °C between the water and ambient air temperatures.

3.3 Protocols

3.3.1 Normobaric hypoxia

Study I involved a protocol of normobaric hypoxic exposure. The protocol consisted of three distinct phases: 1) a control phase consisting of 5-minute breathing normobaric normoxic air (PRE); 2) a 10-minute phase of normobaric hypoxic respiration using a gas mixture containing 10.0% oxygen (EXP); 3) lastly 10-minute phase of normobaric hyperoxic respiration, in which participants breathed 100% oxygen (POST).

3.3.2 Apnea tests

In **Studies II, III, and IV**, a specific apnea test protocol was implemented. The preparation for all the apnea tests stipulated that participants should arrive at the laboratory after 12 hours without

any alcohol consumption or strenuous exercise and 2 hours without eating or drinking caffeine-containing beverages.

Study II included five apneas of maximal duration with face immersion to simulate apneic diving as dry apnea elicits a weaker diving response (Kawakami et al. 1967), and facial immersion induces a similar response as whole body immersion (Moore et al. 1972).

Studies III and IV used a protocol with a static apnea ramp test of 1-min apnea increments to compare physiological changes among participants. Variables such as S_pO_2 , $P_{ET}CO_2$, and $P_{ET}O_2$ exhibit non-linear changes influenced by the apnea duration. The testing procedure comprised three breath-holding periods lasting 1, 2, and 3 minutes, respectively. Before each breath-hold, participants were instructed to fully exhale and then take a deep but not maximal inhalation. This technique typically resulted in an inhalation volume of approximately 85% of their vital lung capacity (Schagatay and Holm 1996).

For all apnea protocols, a 2-minute countdown was used before starting the apneas. At 30 seconds before apnea, a nose clip was applied, and 20 seconds before apnea, a mouthpiece was offered to breathe through. Ten seconds before the apnea, the countdown continued second by second.



Fig. 4. The experimental setup in the laboratory during an apnea test. A pneumatic chest bellows is around the participant's chest. They were carried out by (from left to right) Ph.D. student Eric Mulder, Ph.D. student Frank Pernet, Associate Professor Helen Hanstock, and Doctor Pontus Holmström.

During all the studies, a S_pO_2 safety threshold of 65% was established for stopping hypoxic exposure (Study I) or apnea (Studies II-IV) as a precautionary measure to mitigate the risk of BO. This procedure was never actually activated as all participants interrupted apneas before this level.

3.4 Procedures and measurements

3.4.1 Principles for measuring arterial oxygen saturation.

S_aO_2 is a good indicator of arterial oxygen content, and it's related to arterial P_aO_2 (West and Luks 2021). Its values rely on the principle

that Hb exists in two forms in the blood: Oxygenated (HbO₂) and reduced (Hb). So, S_aO₂ is defined as:

$$S_aO_2 = \frac{HbO_2}{HbO_2 + Hb}$$

(Eq. 3)

It should be remembered that in the numerator could be included other forms such as carboxyhemoglobin, methemoglobin, and sulfhemoglobin. The S_aO₂ read by a pulse oximeter is referred to as S_pO₂. Measuring P_aO₂ and S_aO₂ requires arterial blood sampling. Although pulse oximetry has traditionally been deemed less accurate compared to S_aO₂ (Wilson et al. 2010; Nitzan et al. 2014), recent findings suggest that modern pulse oximeters demonstrate accuracy even in patients with low perfusion conditions (Poorzargar et al. 2022). Thus, measuring S_pO₂ is a non-invasive alternative that has proven reliable even in the demanding environments of intensive care units (Wick et al. 2022) during exposure to acute hypoxia (Burtscher et al. 2004) and at high altitude sojourn (Luks and Swenson 2011).

Pulse oximeters employ two primary technologies for oxygen saturation measurement: transmissive and reflective. These technologies differ in their approach to light detection. Transmissive pulse oximeters rely on a photodetector to capture both red and infrared light. In this design, the emitter and detector are positioned

opposite each other, flanking the tissue assessed (Nitzan et al. 2014). Consequently, transmissive sensors are suitable for use in areas where light can pass through the tissue, such as the fingertip or earlobe. In contrast, reflective pulse oximeters employ backscattered light for measurement. These sensors feature the emitter and detector in the same plane, making them well-suited for placement on flat skin surfaces like the forehead. During **Studies I, II, and IV**, S_pO_2 was measured continuously using transmissive sensors. In **Study III**, a reflectance sensor was placed on the forehead (Table 6).

Table 6. Measurements and equipment used in studies I-IV.

Measurement	Study	Technique	Equipment
S_pO_2	<i>I</i>	Transmissive	Biox 3700e, Ohmeda, Madison, USA
	<i>II</i>	Transmissive	3900, Datex-Ohmeda, Louisville, USA
	<i>III</i>	Reflective	Lifesense, Nonin Medical Inc, Plymouth, USA
	<i>IV</i>	Transmissive	Lifesense, Nonin Medical Inc, Plymouth, USA
$P_{ET}O_2$	<i>I</i>	Paramagnetic based	NormocapOxy, Datex-Ohmeda, Helsinki, Finland
	<i>II</i>	Paramagnetic based	O2100C, Biopac Systems, Inc., Goleta, USA
$P_{ET}CO_2$	<i>I</i>	Infrared based	NormocapOxy, Datex Ohmeda, Helsinki, Finland
	<i>II</i>	Infrared based	CO2100C, Biopac Systems, Inc., Goleta, USA
	<i>III</i>	Infrared based	Lifesense, Nonin Medical Inc, Plymouth, USA
	<i>IV</i>	Infrared based	EMMA capnograph, Masimo Corp, CA, USA
Breathing Movements	<i>II</i>	Pneumatic bellows	Custom made device developed in our laboratory
	<i>III</i>	Force sensor	Prototype developed in our laboratory
Vital Capacity	<i>I</i>	Flow meter	Spirolite 201, Vise Medical Co. Ltd, Chiba, Japan
	<i>II</i>	Flow meter	Compact Expert, Vitalograph, Buckingham, UK
	<i>III</i>	Flow meter	Compact Expert, Vitalograph, Buckingham, UK
	<i>IV</i>	Flow meter	Spirobank II basic, MIR, Rome, Italy
HR	<i>I</i>	Transmissive	Biox 3700e, Ohmeda, Madison, USA
	<i>II</i>	Transmissive	3900, Datex-Ohmeda, Louisville, USA
	<i>III</i>	Reflective	Lifesense, Nonin Medical Inc, Plymouth, USA
	<i>IV</i>	Transmissive	Lifesense, Nonin Medical Inc, Plymouth, USA
MAP	<i>I</i>	Photoplethysmography	Finapress 2300, Ohmeda, Madison, USA
	<i>IV</i>	Electronic Oscillometry	M7, Omron, Kyoto, Japan
Skin blood flow	<i>II</i>	Laser-Doppler flow meter	Periflux 5000, Perimed, Järfälla, Sweden
Hb	<i>I</i>	Photometry	ABx Diagnostics Micros 60 CT, Montpellier, France
	<i>IV</i>	Photometry	HemoCue [Hb] 201+, HemoCue AB, Ängelholm, Sweden
Lactate	<i>IV</i>	Enzymatic detection	Lactate Scout 4, EKF diagnostics, Leipzig, Germany
Spleen dimensions	<i>I</i>	Ultrasound	Mindray DP-6600, Shenzhen Mindray Bio-Medical Electronics CO., Ltd., Shenzhen, China
	<i>II</i>	Ultrasound	MyLab 25 gold equipped with a CA621 probe, Esaote, Genoa, Italy
	<i>IV</i>	Ultrasound	Vscan, GE Healthcare, Illinois, USA, and EagleView, Guangdong, China

3.4.2 Respiratory measurements

In **Studies I** and **II**, the fraction of exhaled end-tidal oxygen was measured with a paramagnetic gas analyzer before and after every apnea (Table 6). The end-tidal pressure of oxygen ($P_{ET}O_2$) was obtained by the product of the fraction and dry barometric pressure (barometric pressure – saturated water vapor pressure). In **all studies**, $P_{ET}CO_2$ (Table 6) served as a surrogate of P_aCO_2 , as the CO_2 arterial to alveolar difference ($P_aCO_2 - P_ACO_2$) is negligible in healthy individuals (Lumb 2021). The difference between $P_{ET}CO_2$ and P_ACO_2 is related to the dead space volume. The range of difference is 2 – 5 mmHg (Satoh et al. 2015). $P_{ET}CO_2$ was measured in **all studies** (Table 6). **Studies I** and **II** measured the fraction of exhaled end-tidal CO_2 . To obtain the partial pressure of CO_2 , the same procedure described for the fraction of exhaled end-tidal oxygen was applied. In **Studies III** and **IV**, the device calculated partial pressure directly. In **Study IV**, the used capnograph reported a difference of 6 mmHg (Kim et al. 2016), which is close to the expected difference. Additionally, the aim of those studies was focused on the changes before and after breath-hold. Despite acknowledged differences, alterations in exhaled CO_2 pressures are expected to follow P_aCO_2 trends. The non-invasiveness and ease of deployment in field studies make this method particularly advantageous.

Respiratory chest movements were continuously monitored in **Studies II and III** (Table 6). Two distinct devices were utilized for this purpose: a laboratory-developed pneumatic chest bellows connected to a pressure sensor with an amplifier (Fig. 4) and a laboratory-developed buckle equipped with force sensors attached to a chest belt and connected to a datalogger (Fig. 3). In **Study IV**, the participants informed the researcher when they had the first IBM by raising one finger of the right hand, and the researcher noted the time.

3.4.3 Cardiovascular measurements

HR was measured continuously via the pulsation signals from the pulse oximeter in all studies. This method has been established as highly accurate, particularly during rest and submaximal exercise (Iyriboz et al. 1991). MAP was continuously assessed using a photoplethysmometer with a finger cuff placed on the right hand in **Study I**. In **Study IV**, baseline measurements of systolic and diastolic blood pressure were taken in triplicate after at least 5 minutes of rest via electronic oscillometry (Table 6). MAP was calculated as $[(\text{systolic blood pressure} + 2 \times \text{diastolic pressure})/3]$. In **Study II**, skin blood flow (SkBF) was quantified through laser Doppler technology and served as a continuous metric for monitoring peripheral vasoconstriction relative to baseline (Table 6).

3.4.4 Hematological measurements

Hb concentration experiences fluctuations due to changes in plasma volume, a phenomenon extensively documented in scenarios like exhaustive exercise (Sjøgaard and Saltin 1982) and high-altitude acute acclimatization (Siebenmann et al. 2023) as water moves out of the intravascular compartment. However, investigations into these variations during voluntary breath-hold events have been less frequent. A recent study (Bailey et al. 2022) demonstrated that breath-holding, coupled with hypercapnia, hypoxemia, and systemic hypertension, can elevate intracranial pressure, disrupting the blood-brain barrier and promoting plasma movement out of capillaries. Nevertheless, as our minimum accepted S_pO_2 was 65%, no participants were expected to reach the value of 60% reported by Bailey et al. (2022). Consequently, no significant variations in plasma volume were anticipated in brief exposures to hypoxic air or during 3-minute breath-holds.

Hb was measured in **Studies I** and **IV** (Table 6). In **Study I**, venous blood samples were collected through a venous catheter placed in the antecubital vein of the left arm. During each exposure (baseline, hypoxia, and hyperoxia), 3 mL of venous blood was collected after inserting rinse and taking a trash sample of 1 mL. These blood samples were promptly analyzed in triplicate (Table 6).

During **Study IV**, capillary blood samples were obtained in triplicate before the protocol initiation and after ten minutes in a prone position. Further capillary blood samples were also obtained within 50-70 seconds after the completion of the third apnea. Each set of blood samples, pre and post apnea, were analyzed immediately after collection. Concurrently, duplicate blood lactate samples were obtained alternating with Hb sampling, and immediate analysis was performed.

3.4.5 Ultrasound spleen measurements

In **Study I**, three axial measurements of splenic length, thickness, and width at one-minute intervals throughout the experimental protocol were performed (Table 6). Baseline measurements started after 20 minutes of sitting rest. In **Study II**, the assessment involved measuring spleen length and thickness at 15-second intervals during the protocol. Baseline measurement started after a 10-minute horizontal rest. As for **Study IV**, after at least 6 minutes of sitting rest, three axial spleen dimensions were measured every minute for at least 5 minutes during the baseline phase. The measurements in **Studies I and IV** were taken with participants in the sitting position, whereas measurements in **Study II** were conducted with participants in the prone position. One skilled researcher conducted all spleen measurements during each of the studies. This methodology has demonstrated high reliability in repeated assessments (Holmström et

al. 2022). The splenic measurements were conducted without blinding, as the dynamic changes and pulsatile characteristics necessitated real-time interpretation. While this approach may potentially introduce bias, it aligns with methodologies used by other research groups that display similar values and responses (Elia et al. 2019, 2021a; Keeler et al. 2022).

3.5 Calculations and data analysis

3.5.1 Continuous measurement analysis

Average baseline values used for S_pO_2 , HR, and SkBF were collected 60-30 seconds before each apnea to minimize effects from the previous apnea and by the anticipation of the following apnea. The S_pO_2 nadir was defined as the lowest S_pO_2 recorded value after each apnea. The lowest S_pO_2 values typically occurred within 30 seconds post apnea due to circulatory delay (Andersson and Schagatay 1998). In a few participants, the S_pO_2 nadir was observed after 60 seconds of resuming breathing. We calculated the average HR during the first 30 seconds of apnea to assess the initial tachycardia during apnea. The mean HR was used to determine the bradycardia during apnea but excluded the first 30 seconds of the apnea. SkBF during apnea was defined as the mean value during the whole duration of apnea. As the measurements are in perfusion units (Fredriksson et al. 2009), the percentage of change during apnea from baseline was used (Schagatay and Andersson 1998).

In **Study IV**, the inflection point in the slope of the S_pO_2 profile was calculated by fitting a piecewise linear model to the S_pO_2 time series data and by computing a likelihood ratio test to identify multiple segments with different statistical properties using the “findchangepts” function in MATLAB (R2023b, MathWorks Inc, Natick, USA), with “linear statistic” option (Killick et al. 2012). The time to reach the inflection point (Fig. 5a) in the slope ($tS\Delta$) was extracted for apnea 2 (A2) and apnea 3 (A3) and was used to determine who had desaturated during A2. The participants were categorized into different groups based on their $tS\Delta$ values. Those who did not exhibit any $tS\Delta$ during A2 were placed in the “No-desaturation” group. Participants who had a measurable $tS\Delta$ during the apnea were classified under the “Fast” group, and those whose $tS\Delta$ was only determined after completing the apnea were categorized under the “Slow” group.

The changes in the HR were calculated by fitting a piecewise linear model to the HR time series data and by computing a likelihood ratio test to identify multiple segments with different statistical properties using the “findchangepts” function in MATLAB, with the “linear statistic” option. Those changes identified the different phases of the diving bradycardia (Fig. 5b).

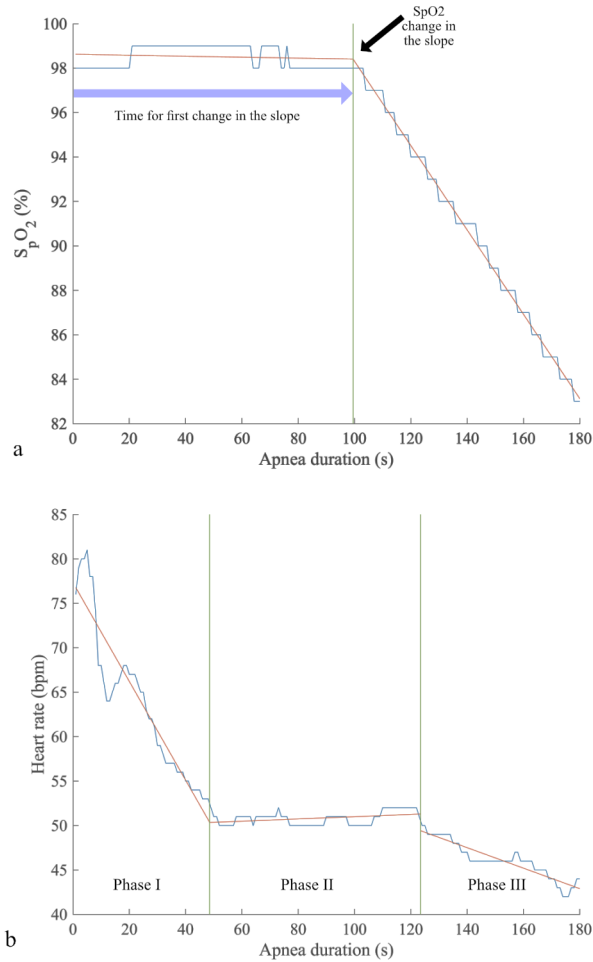


Fig. 5. Calculation of the inflection point of S_pO_2 profile (a) and HR profile (b) in one participant during third apnea. Yellow vertical lines indicate an inflection point in the profile.

3.5.2 Respiratory Measurements

In **Study II**, the onset of IBM was obtained by visually examining the respiratory movements curve. It was defined as the first negative deflection after a flat line, followed by rhythmical negative deflections.

In **Study III**, a force sensor developed in our laboratory (Sieber et al. 2022) was used to estimate tidal volume and measure respiratory rate. Estimating tidal volume through changes in thoracic circumference poses several challenges. Firstly, changes in lung volumes are not entirely reflected in thoracic changes alone, as abdominal circumference changes may be the only way to detect low V_t breathing. Before using it in data collection, a test to evaluate the accuracy of the force sensor was performed in the laboratory (Paper III supplementary information). It included ten adult participants (4 females and 6 males) with a mean $\pm$ SD age of 39 ± 10 years, height 177 ± 9 cm, and weight of 77 ± 8 kg. They conducted a series of six slow vital capacity (SVC) maneuvers, allowing for regular breathing intervals in between. The measurements of SVC were recorded using a spirometer (Compact Expert, Vitalograph, Buckingham, UK) while the participants were seated. The VC did not change across the tests ($p = 0.051$), nor did the amplitude in the signal from the force sensor ($p = 0.123$). The mean VC was 4.6 ± 1.1 L, and the mean amplitude was 6.7 ± 1.8 N. There was a strong correlation between VC and the amplitude ($r_s = 0.712$, $p = 0.021$). A linear regression was used to calculate the predicted VC. The difference between the measured VC and the predicted VC was 0.00 ± 0.7 L. The amplitude and frequency in the last minute before apnea were extracted from the force sensor signal. The number of peaks in 60 seconds represented the respiratory rate, and the signal's amplitude was used as the surrogate

of tidal volume (Monaco and Stefanini 2021). With those values, the estimated minute ventilation (**eMv**) was calculated as the product of the respiratory rate and the amplitude (Fig. 6).

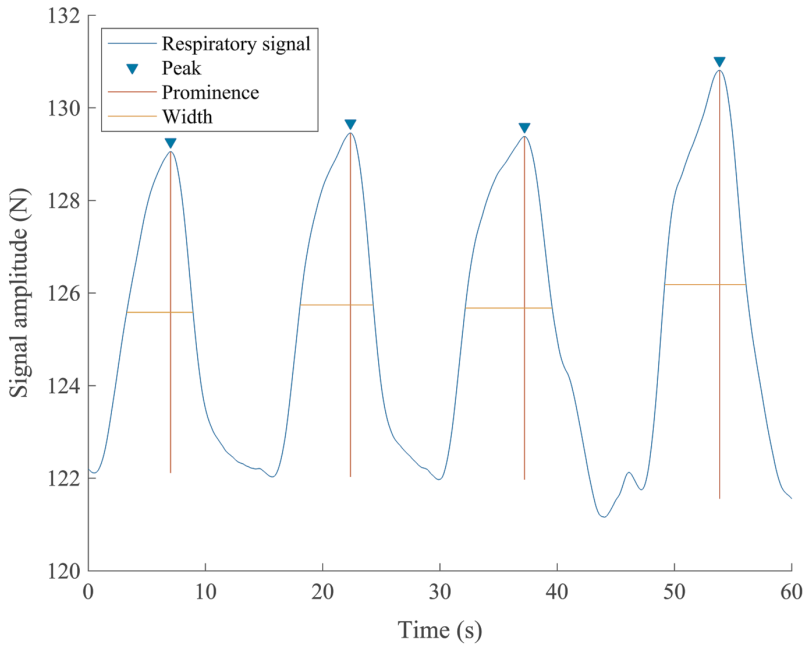


Fig. 6. Representative respiratory signal from one participant 60 seconds before the breath-hold. The inverted triangles show the detected peaks of the signal, corresponding with the end of inhalation. The vertical lines show the amplitude of every wave (Prominence). The horizontal lines show the duration measured at the midpoint of the prominence (Width) N, newtons; s, seconds.

In all studies, P_{ETCO_2} at the start of apnea was defined as the value obtained after the last exhalation before every apnea.

P_{ETCO_2} at the end of apnea was the highest value measured at exhalation directly after apnea.

3.5.3 Spleen volume

In **Studies I** and **IV**, spleen volume was calculated using the Pilström equation.

$$L \times \pi \left(\frac{(W \times T - T^2)}{3} \right) \quad (\text{Eq. 4})$$

using the spleen maximal length (L), width (W), and thickness (T; Fig. 7) obtained from the ultrasound measurements (Schagatay et al. 2005). In **Study I**, PRE, EXP, and POST values were the average of the last three measurements at every stage. In **Study IV**, individual spleen volumes were computed based on the two consecutive highest volumes. In **Study III**, Spleen volume was calculated using length and thickness (Koga and Morikawa 1975)

$$S = 0.8 \times a \times b \quad (\text{Eq.5})$$

where S is the sectional area, a is the length, and b is the thickness. Spleen volume was calculated as a function of the sectional area (Koga 1979)

$$V = (7.53 \times S) - 77.5 \quad (\text{Eq. 6})$$

The baseline was defined as the average of the 5 minutes before the first apnea (20 measurements). During apnea, apnea values were categorized as initial (first recorded), middle (average of measurements), and final (last recorded before breathing resumed). The rationale for using only 2-dimensional measures in **Study II** is related to the short time available when measuring every 15 seconds. Splenic contraction refers to the change in spleen volume from baseline measurements, expressed in both percentage and absolute values.

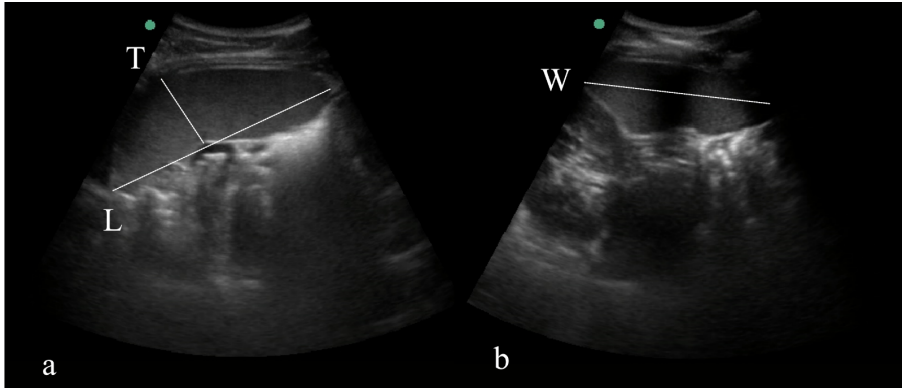


Fig. 7. Splenic measurements of (a) maximal length (L), thickness at the hilum (T), and (b) width (W).

3.6 Statistical analysis

All statistical analysis was performed using the software SPSS 26 and 27 (IBM Corp, Armonk, USA) with a level of significance set at $p < 0.05$. All studies' data were tested for normality using Shapiro-Wilk's test. All data is presented as mean $\pm$ standard deviation (SD). In

Study I, the Paired Student's t-test was used to compare PRE and EXP gas values. Assessments of interaction effects between the independent variables (treatment [PRE, EXP; POST] and sex on the dependent variables (spleen volume, Hb, S_pO_2 , and HR) were conducted by a two-way mixed analysis of variance (ANOVA) with Bonferroni corrections for repeated measures. Assessments of associations between dependent variables were conducted with bivariate correlation tests using Pearson product-moment correlation coefficient (r). In **Study II**, a two-factor [time (apnea 1 to apnea 5) vs. group (normal breathing and hyperventilation)] repeated measures ANOVA was used to compare apnea duration, $P_{ET}O_2$, $P_{ET}CO_2$, S_pO_2 , HR, IBM-onset, and splenic contraction. In **Study III**, Spearman's correlation was performed to assess the relationship between the VC and $P_{ET}CO_2$ with the amplitude of the signal from the force sensor. A one-way repeated measures ANOVA was used to compare the VC and amplitude of the respiratory signal and to compare $P_{ET}CO_2$, amplitude, and eMv before every apnea. A linear regression was performed to predict the VC from the amplitude of the signal and $P_{ET}CO_2$ from the eMV before every apnea. The Bland-Altman method (Bland and Altman 1986) was used to assess the agreement between measured VC and predicted VC and between $P_{ET}CO_2$ and predicted $P_{ET}CO_2$. In **Study IV**, a two-factor [time (apnea 1 to apnea 3) vs. group (male and female)] repeated measures ANOVA was used to compare age, height, weight, body mass index (BMI), vital capacity, spleen

volume, hemoglobin, lactate, and blood pressure. A two-factor[time (baseline, A1-A3, and recovery vs. group (*No-desaturation, Slow, Fast*))] was used to compare age, height, weight, BMI, vital capacity, spleen volume, hemoglobin, lactate, blood pressure, S_pO_2 , HR, $P_{ET}CO_2$. A two-factor [time (30 s samples) vs. group (*No-desaturation, Slow, Fast*)] was used to compare S_pO_2 and HR. In **Studies II** and **IV**, when a two-way interaction was found, a one-way ANOVA was performed to evaluate the simple main effects of conditions and apnea. A Bonferroni correction was applied for multiple comparisons. In **Study I**, meaningfulness of effects was estimated by the standardized mean difference (Cohen's d, effect size [ES]) computed as the mean difference divided by the pooled SD. ES was presented along with 95% confidence intervals [CI]. Interpretation of ES was based on three categories: 0.0–0.5 was considered a small effect; 0.6–1.1 a medium effect; and ≥ 1.2 a large effect. In **Studies II, III**, and **IV**, ES were estimated by the partial eta squared (η_p^2) and the generalized eta squared (η_G^2) and are presented with a 90% CI. An effect size of 0.01 – 0.05 was considered small, from 0.06 – 0.13 was considered medium, and from and above 0.14 was considered large (Cohen 1988; Bakeman 2005; Lakens 2013).

4 Results

4.1 Laboratory studies

4.1.1 Study I – Splenic contraction and hypoxia

Exhaled gases and oxygen saturation

In Study I, 15 participants were exposed to normobaric hypoxia (10% oxygen) for 10 minutes, followed by 10 minutes of breathing 100% oxygen. During the normobaric hypoxic exposure, the $P_{ET}O_2$ decreased from the baseline values of 98.6 ± 17.2 mmHg to 45.5 ± 6.3 mmHg ($p < 0.001$; ES = 4.00 [3.26 – 4.75]) and the $P_{ET}CO_2$ decreased from 32.7 ± 4.3 mmHg to 28.3 ± 4.7 mmHg ($p < 0.001$; ES = 0.98 [0.23 – 1.72]). Baseline S_pO_2 was 96.4 ± 1.7 % and dropped to 74.7 ± 8.4 % by the end of the normobaric hypoxic exposure ($p < 0.001$; ES = 3.94 [3.20 – 4.69]; Fig. 8).

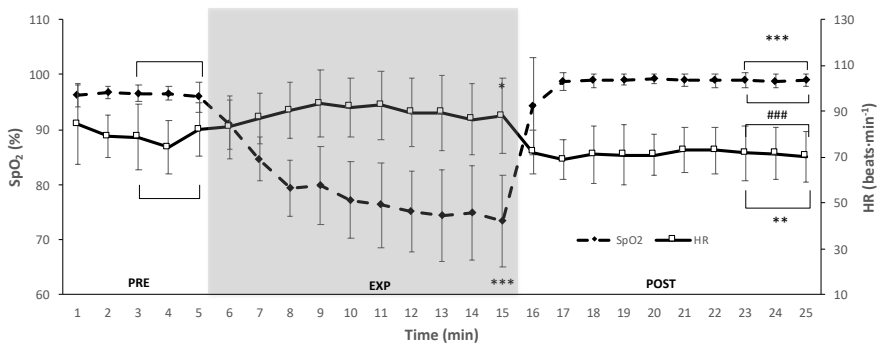


Fig. 8. Peripheral arterial oxygen saturation (S_pO_2) and heart rate (HR) during baseline 5 min normoxic breathing (PRE), 10 min hypoxic breathing of 10 % O_2 (EXP), and 10 min hyperoxic breathing of 100 % O_2 (POST). The grey zone indicates EXP. Horizontal brackets indicate an average value for that period. * indicate $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$ compared with PRE. ### indicate $p < 0.001$ compared with EXP. $n = 15$

Splenic contraction and hemoglobin concentration

One participant's spleen dimensions weren't measured due to its unusual shape, which didn't meet the formula requirements. Thus, 14 participants were included. After 3 minutes of hypoxic exposure, the spleen volume decreased from a mean baseline of 279 ± 115 mL to 219 ± 75 mL ($p = 0.005$; ES = 0.62 [-0.16 – 1.39]; Fig. 9). During the last 3 minutes of the 10-minute hypoxic exposure, the spleen volume reduced further to 201 ± 93 mL ($p < 0.001$; ES = 0.75 [-0.03 – 1.52]; Fig. 9). At the end of the hyperoxic exposure, the spleen volume returned to the baseline values (279 ± 120 mL ($p = 1.000$; ES = 0.00 [-0.78 – 0.78]; Fig. 9).

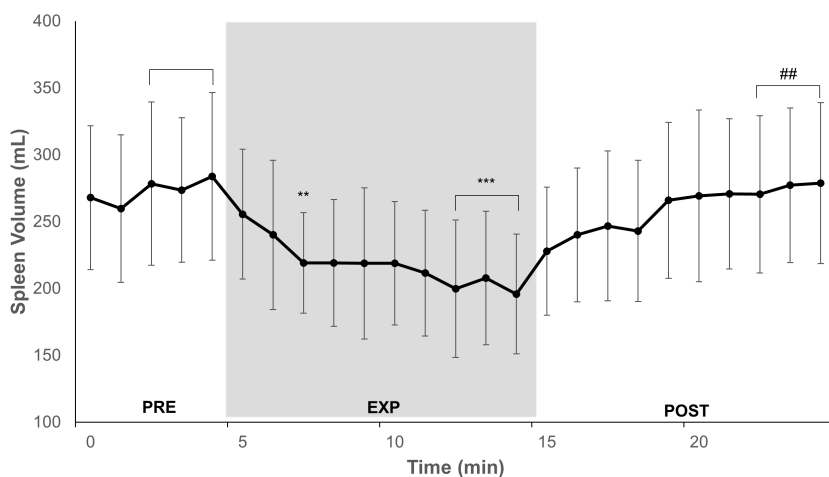


Fig. 9. Spleen volume during baseline 5 min normoxic breathing (PRE), 10 min normobaric hypoxic breathing of 10 % O₂ (EXP), and 10 min hyperoxic breathing of 100 % O₂ (POST). Grey zone indicates EXP. Horizontal brackets indicate an average value for that period. ** indicate $p < 0.01$ and *** $p < 0.001$ compared to PRE. ## indicates $p < 0.01$ compared to EXP. $n = 14$

In one participant, we were unable to obtain an Hb sample, so only 14 participants were included in the analysis. The mean baseline Hb was 138.8 ± 7.6 g/L, and it increased after the normobaric hypoxic exposure to 142.9 ± 8.1 g/L ($p < 0.001$; ES = 0.52 [-0.26 – 1.30]). At the end of the hyperoxic exposure, the Hb returned to baseline values (140.2 ± 8.0 ($p = 0.734$; ES = 0.18 [-0.60 – 0.96])). There was a strong correlation between the change in S_pO_2 from PRE to EXP and the change in Hb from pre to EXP ($r = 0.7$, $p = 0.007$). There was, however, no correlation between splenic volume reduction from PRE to EXP and the change in Hb from pre to EXP ($r = 0.1$, $p = 0.736$).

4.1.2 Study II – Effects of hyperventilation during serial apneas.

Hyperventilation and S_pO_2

In Study II, 18 participants performed two series of five apneas of maximal duration separated with a two-minute rest in between. In one of the series, apnea was preceded by normal breathing (*NB*), while the other series of apnea was preceded by 15 seconds of hyperventilation (*HV*). The mean baseline S_pO_2 was similar between conditions, *NB* and *HV*, and remained unchanged from apnea 1 - 5 ($p = 0.421$, $\eta_p^2 = 0.04$, 90% CI [0.00 – 0.24], $\eta_G^2 = 0.01$; Fig. 10a). S_pO_2 nadir did not change across the apnea series in *NB* ($p = 0.576$, $\eta_p^2 = 0.03$, 90% CI [0.00 – 0.14], $\eta_G^2 = 0.10$, Fig. 10b) but decreased during the apnea series in *HV* (Fig. 10b) with the lowest value in A5 ($p < 0.001$, $\eta_p^2 = 0.48$, 90% CI [0.25 – 0.61], $\eta_G^2 = 0.17$). Mean S_pO_2 nadir was lower in *HV*

compared with *NB* (90.6 ± 5.6 %, vs. 93.6 ± 4.9 %, respectively, $p < 0.001$, $\eta_p^2 = 0.644$, 90% CI [0.35 – 0.76], $\eta_G^2 = 0.10$; Fig. 10b). In four participants, the S_pO_2 nadir dropped below 75% during the fifth apnea in the *HV* trial.

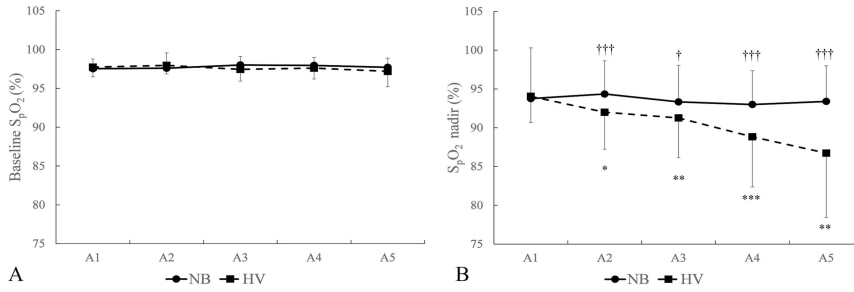


Fig. 10. Mean baseline pre-apnea S_pO_2 from A1-A5 (A) and at post-apnea nadir (B). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ compared with A1 in the same condition; † $p < 0.05$, †† $p < 0.01$, ††† $p < 0.001$ compared with HV. $n = 18$

Hyperventilation and the diving response

The mean HR at baseline was similar in both conditions (*NB* and *HV*), with a mean value of 68 ± 12 bpm in *NB* and 68 ± 9 bpm in *HV* ($p = 0.923$, $\eta_p^2 = 0.001$, 90% CI [0.00 – 0.04], $\eta_G^2 = 0.0001$; Fig. 11a). Baseline HR was unchanged between apneas ($p = 0.795$, $\eta_p^2 = 0.02$, 90% CI [0.00 – 0.15], $\eta_G^2 = 0.004$; Fig. 11a).

The mean HR during the first 30 s of apnea (HR_{peak}) was higher in *HV* compared to *NB* (80 ± 15 bpm and 66 ± 13 bpm, respectively, $p < 0.001$, $\eta_p^2 = 0.76$, 90% CI [0.54 – 0.84], $\eta_G^2 = 0.23$; Fig. 11b). During the *NB* series, HR_{peak} decreased from 71 ± 14 bpm in A1 to 62 ± 11 bpm in A5 ($p < 0.001$, $\eta_p^2 = 0.39$, 90% CI [0.21 – 0.48], $\eta_G^2 = 0.07$; Fig. 11b).

Across the *HV* series, HR_{peak} dropped from A1, at 87 ± 17 bpm, to A5 at 75 ± 13 bpm, ($p < 0.001$, $\eta_p^2 = 0.36$, 90% CI [0.18 – 0.45], $\eta_G^2 = 0.07$; Fig. 11b).

The mean HR during apnea was similar between *NB* and *HV* (54 ± 7 bpm and 55 ± 8 bpm, $p = 0.205$, $\eta_p^2 = 0.09$, 90% CI [0.00 – 0.32], $\eta_G^2 = 0.01$; Fig. 11c). Neither were there any differences within the groups in mean HR during the repeated apneas ($p = 0.232$, $\eta_p^2 = 0.08$, 90% CI [0.00 – 0.19], $\eta_G^2 = 0.01$; Fig. 11c). The HR reduction in percent from baseline was similar for both *NB* and *HV* (19% and 18%, respectively, $p = 0.694$, $\eta_p^2 = 0.09$, 90% CI [0.00 – 0.17], $\eta_G^2 = 0.003$).

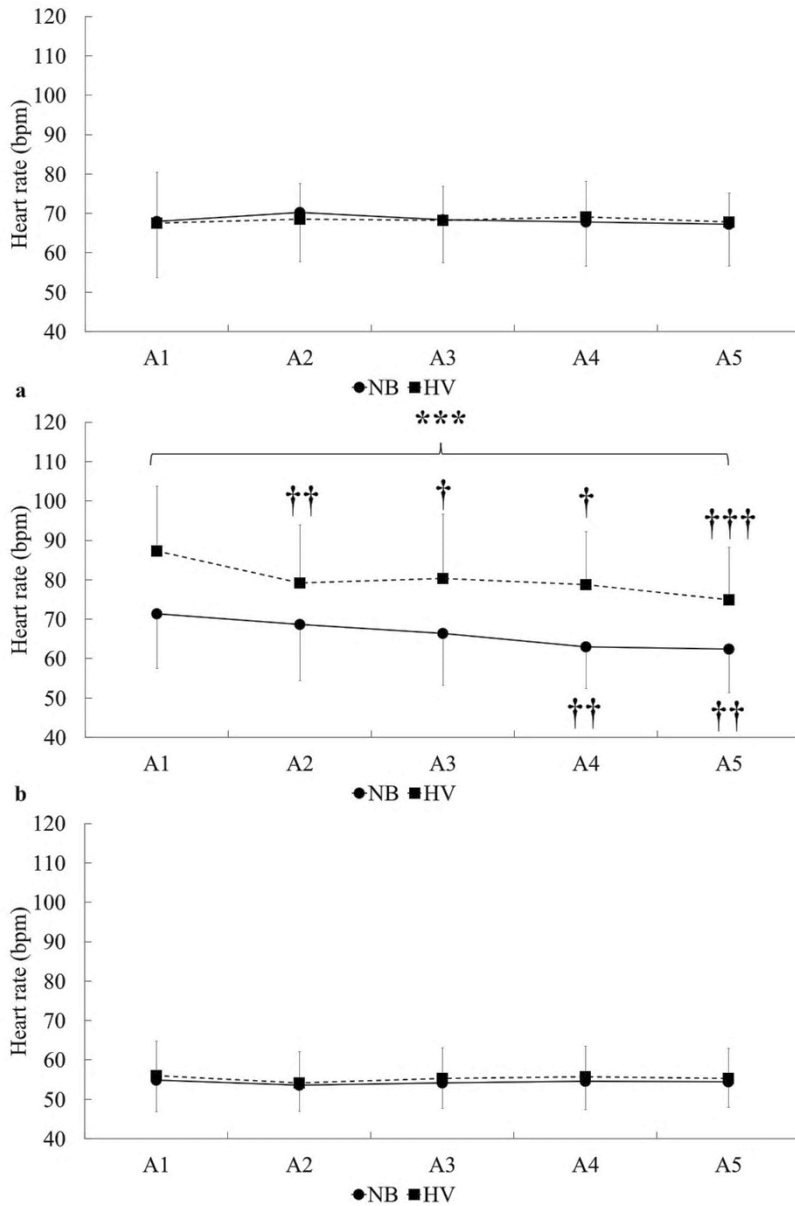


Fig. 11. Heart rate average for baseline (a), HR_{peak} (b), and during apnea (c). *** $p < 0.001$ between normal (NB) and hyperventilation (HV) conditions; ††† $p < 0.001$, †† $p < 0.01$, and † $p < 0.05$ compared with A1 in the same condition.

There was no difference in the SkBF reduction between *NB* and *HV* ($-10.0 \pm 17.3\%$ and $-8.1 \pm 18.3\%$, $p = 0.10$, $\eta_p^2 = 0.10$, 90% CI [0.00 – 0.33], $\eta_G^2 = 0.01$). Furthermore, there were no differences observed between apneas within the same group ($p = 0.716$, $\eta_p^2 = 0.02$, 90% CI [0.00 – 0.10], $\eta_G^2 = 0.01$).

Hyperventilation and splenic contraction

Mean baseline spleen volume was similar between conditions, with 197 ± 54 mL in *NB* and 194 ± 56 mL in *HV* ($p = 0.276$). During apneas, mean spleen volume was 131 ± 40 mL in *NB* and 126 ± 43 mL in *HV* series ($p = 0.348$), reflecting reductions from baseline by 33% and 35%, respectively ($p < 0.001$). The mean spleen volume during the two minutes of recovery was similar at 176 ± 46 mL in *NB* and 179 ± 57 mL in the *HV* series ($p = 0.710$). The average spleen volume during recovery after every apnea was still 11% lower than baseline in *NB* and 8% lower in *HV* ($p = 0.012$). After 5 minutes of recovery, spleen volume was similar to the baseline value with 195 ± 53 in *NB* and 192 ± 55 in *HV* ($p = 0.620$), with no difference between conditions ($p = 0.452$).

Hyperventilation and apnea breaking points

The end-tidal gas variables ($P_{ET}O_2$ and $P_{ET}CO_2$) were initially measured on 18 participants, but only 13 participants' data was acquired due to gas measurement module dysfunction. At the start of each apnea, $P_{ET}CO_2$ was lower in *HV* (17.4 ± 3.4 mmHg, 2.3 ± 0.5 kPa)

compared to *NB* (29.0 ± 4.0 mmHg, 3.9 ± 0.5 kPa, $p < 0.001$, $\eta_p^2 = 0.93$, 90% CI [0.86 – 0.96], $\eta_G^2 = 0.76$; Fig. 12). $P_{ET}CO_2$ was lower at the start of the last apnea in both conditions compared with the first ($p = 0.001$, $\eta_p^2 = 0.53$, 90% CI [0.32 – 0.61], $\eta_G^2 = 0.11$; Fig. 12). At the end of apneas, the mean $P_{ET}CO_2$ was lower in *HV* with 38.5 ± 5.2 mmHg (5.1 ± 0.7 kPa), compared to *NB* with 40.3 ± 5.5 mmHg (5.4 ± 0.7 kPa, $p = 0.005$, $\eta_p^2 = 0.49$, 90% CI [0.12 – 0.67], $\eta_G^2 = 0.03$; Fig. 12). $P_{ET}CO_2$ at the end of apnea was similar across the apnea series ($p = 0.272$, $\eta_p^2 = 0.10$, 90% CI [0.00 – 0.25], $\eta_G^2 = 0.006$; Fig. 12).

$P_{ET}O_2$ was higher at the start of each apnea in *HV* with 137.2 ± 7.4 mmHg (18.0 ± 1.0 kPa) compared to *NB* with 121.6 ± 11.1 mmHg (16.2 ± 1.5 kPa, $p < 0.001$, $\eta_p^2 = 0.917$, 90% CI [0.80 – 0.94], $\eta_G^2 = 0.43$; Fig. 12). Starting $P_{ET}O_2$ remained unchanged from A1 to A5 in both conditions ($p = 0.094$, $\eta_p^2 = 0.25$, 90% CI [0.05 – 0.35], $\eta_G^2 = 0.009$; Fig. 12). $P_{ET}O_2$ at the end of each apnea was lower in *HV* with a mean of 83.4 ± 17.3 mmHg (11.1 ± 2.3 kPa), compared to *NB* with 89.4 ± 15.6 mmHg (11.9 ± 2.1 kPa, $p = 0.005$, $\eta_p^2 = 0.498$, 90% CI [0.12 – 0.68], $\eta_G^2 = 0.04$; Fig. 12). The $P_{ET}O_2$ values at the end of apnea in *HV* decreased throughout from A1 to A5 ($p < 0.001$, $\eta_p^2 = 0.426$, 90% CI [0.16 – 0.55], $\eta_G^2 = 0.03$; Fig. 12).

Only 7 participants had IBM in all the apneas and in both conditions. Overall, the first IBM was delayed in *HV* by 112 ± 49 seconds, compared with 89 ± 42 seconds in *NB* ($p = 0.003$).

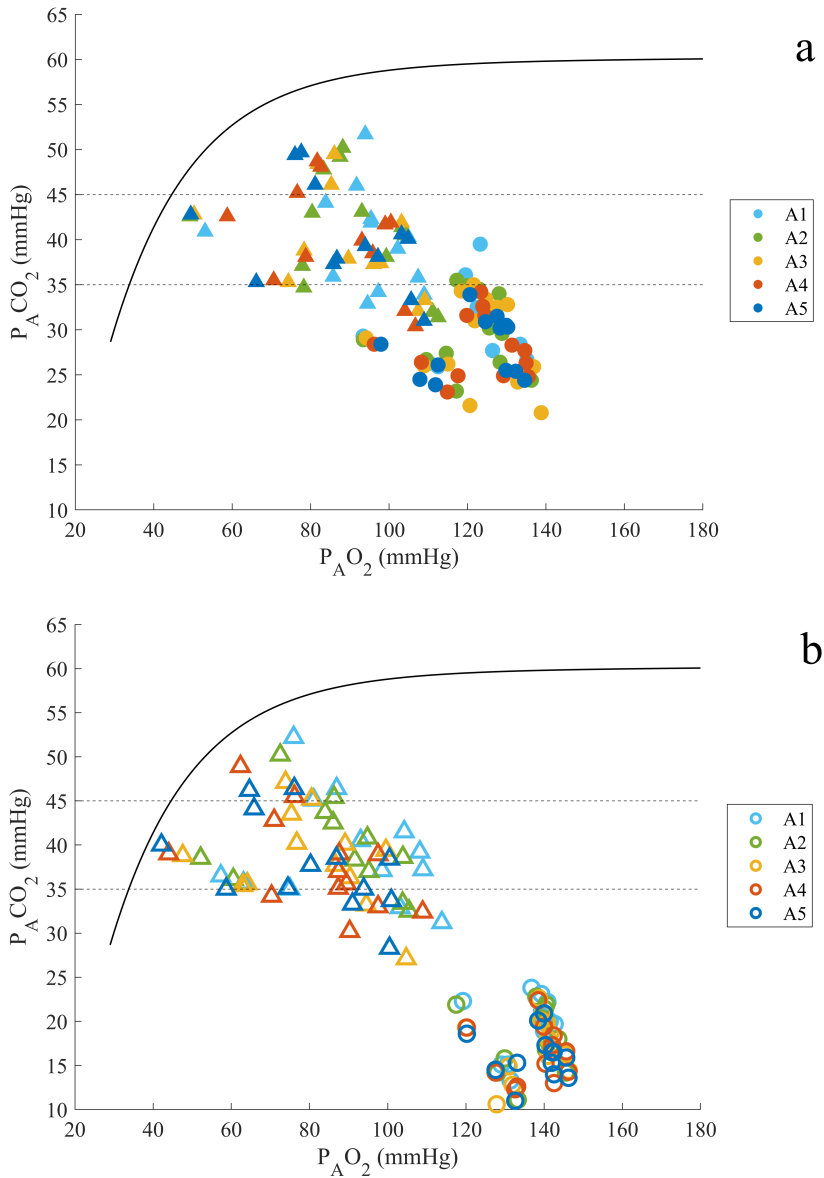


Fig. 12. O₂-CO₂ diagram showing individual alveolar values during *NB* (a) and *HV* (b). Circles represent values at the start of apnea. Triangles represent values at the end of apnea. Filled symbols indicate *NB* (a), and empty symbols indicate *HV* (b). Colors show the different apnea series (A1 – A5). Horizontal dashed lines show the range of normal P_ACO₂. Black continuous lines represent the conventional apnea breaking point (Agostoni 1963). n =13.

4.2 Field laboratory studies

4.2.1 Study III – Estimating Hyperventilation in Freedivers

In Study III, 21 participants performed a static apnea ramp protocol using three apneas of 1, 2, and 3-minute duration with a 2-minute rest period between apneas in a sitting position while cardiorespiratory variables were recorded. $P_{ET}CO_2$ was measured before and after each apnea. The frequency and amplitude of chest movements were recorded using the laboratory-constructed force sensor attached to a chest belt. The product of respiratory rate and signal amplitude (eMv) before every apnea was used as a proxy of minute ventilation. Hyperventilation before every apnea was identified by reduced values of $P_{ET}CO_2$ (30.5 ± 5.3 , 31.3 ± 6.6 , and 29.3 ± 6.4 mmHg, respectively).

End-tidal CO_2 prediction

A linear regression was used to establish the relationship between $P_{ET}CO_2$ and eMV for every apnea (Fig. 13a, 13b, 13c) and further used to predict $P_{ET}CO_2$ with eMV values before each apnea.

The linear regression analysis revealed a statistically significant model for A1($F(1,18)= 6.63$, $p = 0.019$), A2($F(1,19)= 13.99$, $p = 0.001$), and A3($F(1,19)= 9.60$, $p = 0.006$). With an adjusted R^2 for A1-A3 of 0.23, 0.4 and 0.30 respectively. The standard error of the estimate for

A1-A3 was 4.6, 5.1, and 5.4, respectively. A Bland – Altman plot analysis showed that the difference between measured $P_{\text{ET}}\text{CO}_2$ and predicted $P_{\text{ET}}\text{CO}_2$ was -0.00 ± 4.51 mmHg for A1, -1.15 ± 5.04 mmHg for A2, and -0.00 ± 5.24 mmHg for A3 (Fig. 13d, 13e, 13f).

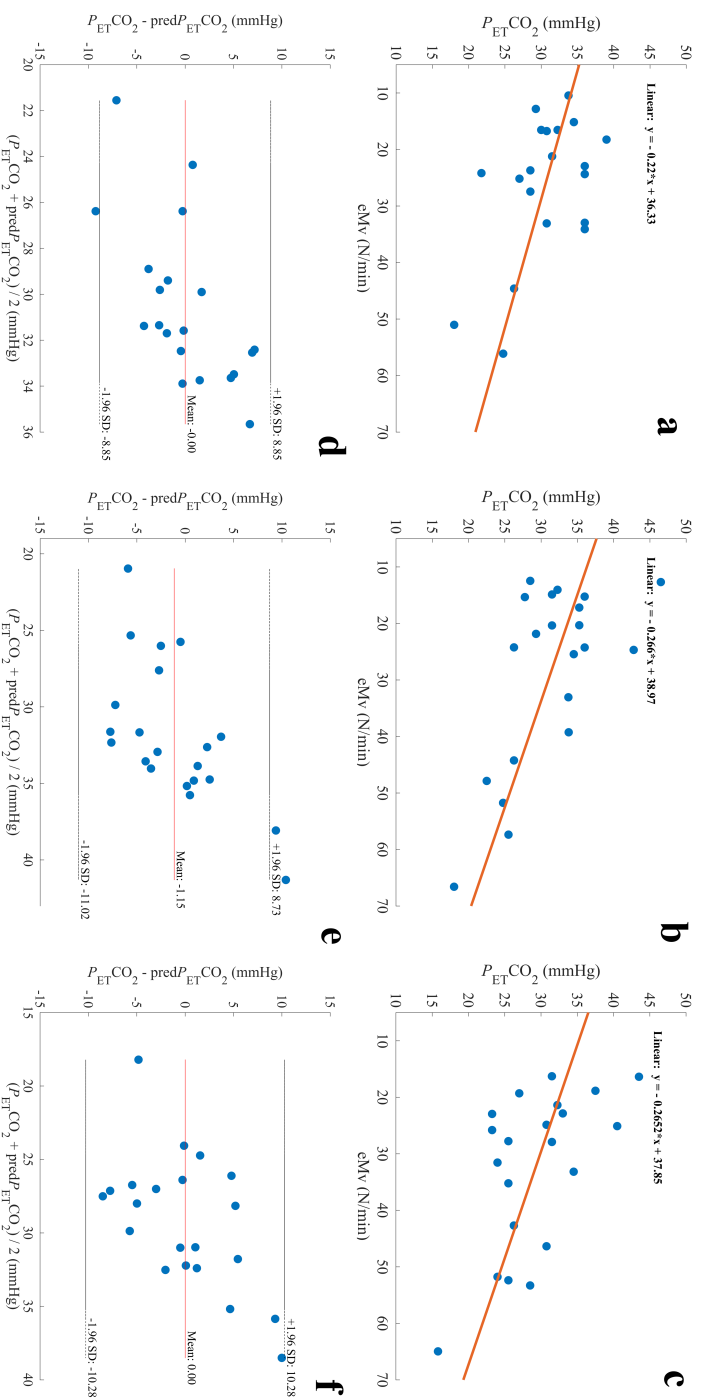


Fig. 13. Comparison of measured end-tidal CO₂ ($P_{ET}CO_2$) with the estimated minute ventilation (eMV) before A1 (a), A2 (b) and A3 (c) and Bland-Altman plots of the difference between $P_{ET}CO_2$ and $predP_{ET}CO_2$ before A1 (d), A2 (e) and A3 (f). The orange line represents the regression line, and the corresponding formula is expressed on each graph. The dotted lines represent the upper limit of agreement

4.2.2 Study IV – oxygen desaturation rate.

Desaturation groups and oxygenation variables

In study IV, 93 participants were recruited, and a static apnea ramp test was performed comprising three apneas of increasing duration (1, 2, and 3 minutes interspersed with a 2-minute rest period). In the 1st apnea (A1), only 10 participants had a determinable inflection point time ($tS\Delta$). During the 2nd apnea (A2), using a 2 min duration, 29 participants did not show a determinable inflection point time $tS\Delta$ in the S_pO_2 curve (*No-desaturation*), 32 participants had a $tS\Delta$ longer than apnea time (*Slow*), and 32 participants had a $tS\Delta$ lower or equal to apnea time (*Fast*). Thus, the participants were categorized into three groups (*No-desaturation*, *Slow*, and *Fast*).

The participants within the *No-desaturation* group exhibited a lower BMI and a larger spleen volume compared to participants in the other two groups (Table. 7).

Table 7. Anthropometrics and baseline values among groups with different desaturation profile in A2.

	No-desaturation (<i>n</i> = 29)	Slow (<i>n</i> = 32)	Fast (<i>n</i> = 32)
Age (years)	35 ± 8	37 ± 8	39 ± 10
Male/female (<i>n/n</i>)	19/10 (34% women)	18/14 (44% women)	18/14 (44% women)
Category (4-5)	25 (86%)	27 (84%)	25 (78%)
Training (hr/w)	8 ± 5	8 ± 7	8 ± 7
Height (cm)	177 ± 9	174 ± 8	174 ± 12
Body mass (kg)	67 ± 11	71 ± 12	72 ± 15
BMI (kg/m ²)	21 ± 2	23 ± 3*	23 ± 3*
Vital capacity (mL/cm)	34.3 ± 7.4	32.5 ± 5.3	32.9 ± 8.0
Spleen volume (mL/cm)	1.34 ± 0.46 (<i>n</i> = 28)	1.08 ± 0.36* (<i>n</i> = 31)	1.04 ± 0.26* (<i>n</i> = 28)
MAP (mmHg)	93 ± 8	91 ± 11	94 ± 14
STA PB (s)	357 ± 94 (<i>n</i> = 28)	302 ± 68*	305 ± 69* (<i>n</i> = 30)

w, weeks. Vital capacity and spleen volume are scaled to the participant's height. Italic numbers in parentheses indicate different numbers of participants. * Significantly different from the *No-desaturation* group.

In the first apnea (A1), the *Fast* group was the only one that showed a reduction in S_pO_2 , from baseline at $97.6 \pm 1.1\%$ to a nadir of $96.8 \pm 1.5\%$ ($p < 0.05$; Table 8). After A2, the *Fast* group showed the lowest S_pO_2 with $93.1 \pm 2.2\%$ compared with the *Slow* and *No-desaturation* groups (94.7 ± 1.3 and $97.4 \pm 0.9\%$, respectively, $p < 0.05$; Table 8). The S_pO_2 nadir was lower during the longest apnea (A3) in all groups ($p < 0.001$, $\eta_p^2 = 0.84$, 90% CI [0.80 – 0.86], $\eta_c^2 = 0.74$; Table 8). During A1, the *Fast* group was the only group that had a reduction in S_pO_2 nadir

compared with the baseline (Table 8). During A2, the *Fast* group had the lowest S_pO_2 . In the *No-desaturation* group, all participants had a S_pO_2 nadir over 95% (Table 8). The *Fast* group had the lowest S_pO_2 nadir during A3, followed by the *Slow* group, and the *No-desaturation* group had the highest value ($p < 0.001$, $\eta_p^2 = 0.34$, 90% CI [0.22 – 0.42], $\eta_G^2 = 0.21$; Table 8).

Table 8. Comparative Analysis of Oxygen Saturation During Apnea Testing Across Different Groups

		<i>No-desaturation</i>	<i>Slow</i>	<i>Fast</i>
<i>n</i>		29	32	32
	BL	98.1 ± 0.9	97.7 ± 1.1	97.6 ± 1.1
	A1	97.8 ± 0.8	97.3 ± 1.3	96.8 ± 1.5 [#]
S_pO_2	R1	98.8 ± 0.6	98.2 ± 0.8	98.0 ± 1.0
(%)	A2	97.4 ± 0.9 [†]	94.7 ± 1.3 [†]	93.1 ± 2.2 [†]
	R2	98.5 ± 0.7	97.9 ± 1.0	98.0 ± 0.9
	A3	93.3 ± 0.7 [†]	89.9 ± 2.2 [†]	86.9 ± 4.0 [†]
	R3	98.3 ± 0.9	97.6 ± 1.2 [#]	97.4 ± 1.2 [#]

BL is the baseline mean value; A1-A3 is the mean nadir value during the respective apnea; R1-R3 is the mean value after each apnea. * Significantly different from all other values in the same group. † Significantly different from the other groups. # Significantly different from the *No-desaturation* group

Desaturation groups characteristics

HR nadir was lower than baseline in all apneas ($p < 0.001$, $\eta_p^2 = 0.71$, 90% CI [0.66 – 0.75], $\eta_G^2 = 0.49$). However, the HR nadir did not change in the *No-desaturation* group ($p = 0.159$), but it decreased with every apnea in the other two groups ($p < 0.001$). During A3, after the initial HR drop, a second drop was observed (third phase) for the *Slow* at 129 s and for the *Fast* at 124 s of apnea (Fig. 14a). The HR nadir at the

end of apnea was lower than the 160-s values for the *Slow* ($p = 0.010$; Fig. 14b) and *Fast* ($p < 0.001$; Fig. 14b) groups. The *No-desaturation* group did not show a third phase in HR (Fig. 14a), and the final values were similar to the 160-s values ($p = 0.861$; Fig. 14b).

The time for the first IBM in A3 was similar among the groups, with 115 ± 23 , 110 ± 26 , and 107 ± 25 seconds for the *No-desaturation*, *Slow* and *Fast*, respectively ($p = 0.872$).

Hb values increased from 137.1 ± 13.2 g/L to 140.8 ± 12.6 g/L after apneas ($p < 0.001$). The post-apnea increase was not significant only in the *No-desaturation* group ($p = 0.052$).

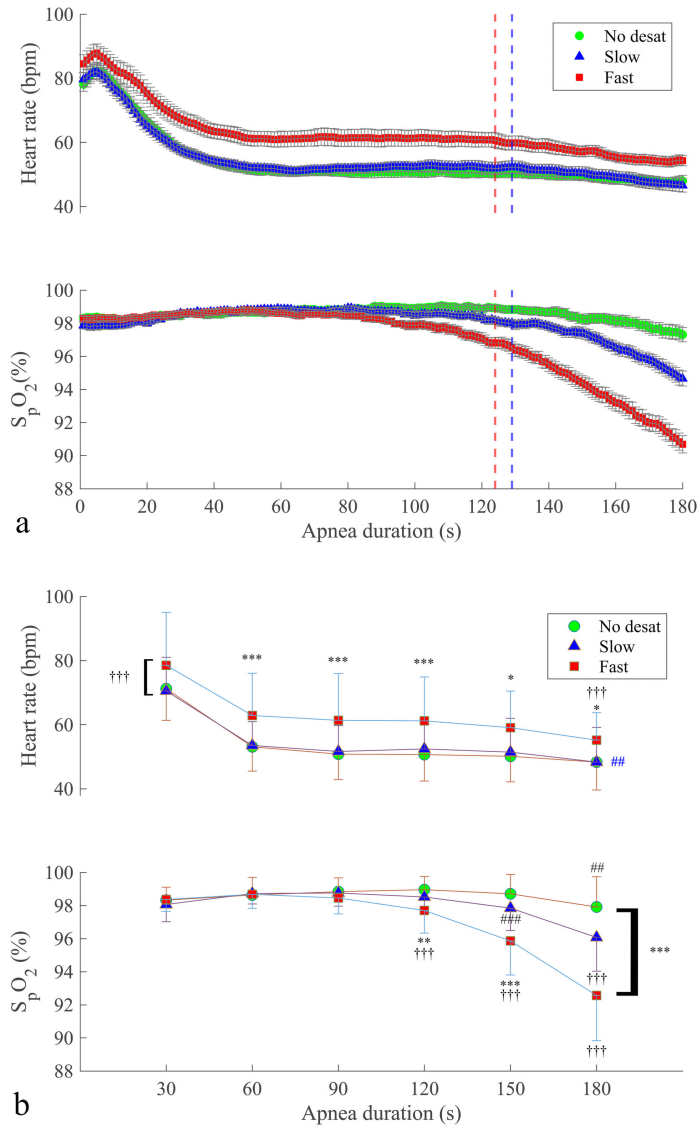


Figure 14. HR and SpO₂ time course (a) and 30 s mean (b) during A3. Green circles represent data from the *No-desaturation* group, blue triangles represent data from the *Slow* group, and red squares represent data from the *Fast* group. The dashed vertical lines (a) indicate the initiation of the third phase on HR, red for the *Fast* group and blue for the *Slow* group. * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$ compared to other groups; ††† $p < 0.001$ compared with other values in the same group; ## $p < 0.01$ (*Slow*), ### $p < 0.001$ compared with previous value in the same group. Data are presented as mean $\pm$ SE (a) and mean $\pm$ SD (b).

4.3 Summary of results

Acute normobaric hypoxia induces a rapid reduction in spleen volume, with a 21% decrease in the first 3 minutes and a maximum contraction of 28% after 10 minutes (**Study I**). Therefore, the main effect is noticeable after 3 minutes.

Serial consecutive apneas, when preceded by hyperventilation, lead to prolonged apnea duration and more severe oxygen desaturation. Despite the progressively increased hypoxemia, these consecutive apneas do not show an augmented diving response or splenic contraction (**Study II**).

Unnoticed hyperventilation is common among freedivers, primarily resulting from increased tidal volume (V_t); it is possible to identify hyperventilation from the signal of a force sensor attached to a chest belt (**Study III**).

The rate of oxygen desaturation varies greatly among individuals, and conducting a static apnea ramp test allows the identification of those who are at a higher risk of developing severe desaturation, which could result in a BO in actual dives. Particularly, individuals with smaller spleen volume and reduced diving bradycardia exhibited more significant oxygen desaturation (**Study IV**).

5 Discussion

The main objective of this thesis was to examine the physiological processes that protect the body during freediving against hypoxia. It aimed to understand how hyperventilation, a known risk factor, could impact these protective mechanisms, particularly in repetitive dives, and how its impact could increase the risk of hypoxic BO.

Study I explored the spleen contraction, which, by expelling stored red blood cells, increases the available blood oxygen during freediving. It has shown that this response may act as a defense against hypoxia within only 3 minutes of exposure and that, contrary to previously believed (Schagatay et al. 2001), it could also develop during a single long dive if hypoxemia develops. This was confirmed by **Study II**, involving serial apneas, where the spleen contracted to the same extent across all apneas, and in **Study IV**, it was confirmed that a large spleen is characteristic of participants less prone to hypoxemia. **Study II** also showed that when serial dives lead to progressively increased apneic duration and oxygen desaturation, especially after hyperventilation, the spleen contraction does not increase to counteract this effect. In **Study II**, the cardiovascular diving response was also studied. However, this response did not compensate for the increased apneic duration and oxygen desaturation observed during hyperventilation. Thus, hyperventilation before serial diving poses a risk to the freediver and

should be avoided. However, this has proven to be difficult as hyperventilation occurred despite instructions to avoid it in **Studies II, III, and IV**.

One way to warn a diver of hyperventilation is to monitor their exhaled CO₂ levels using a CO₂ monitor attached to their snorkel. However, this method has proven challenging because CO₂ sensors are sensitive to wet and moist environments. Moreover, the freediver may bring the snorkel underwater between dives, and most competitive freedivers do not use snorkels at all. To approach this problem, **Study III** explored the potential of detecting hyperventilation before three consecutive apneas by recording the respiratory pattern using a force sensor attached to a chest belt. This method holds potential, but further development is necessary to detect decreasing CO₂ levels between apneas reliably. This technique can be used underwater, and additional research is needed to determine how it can be enhanced and applied to actual freediving.

In **Study IV**, the individual desaturation rate was found to be associated with the diving response and the spleen volume. This is in line with previous findings that these responses are indeed protective. A person with a weak diving response and a small spleen could be more vulnerable to BO as they could desaturate faster. Below, I will discuss each of these findings in more detail.

5.1 Protective Mechanisms

5.1.1 Diving response

In **Study IV**, a static apnea ramp test, with fixed apneic durations, was used to allow the comparison of the oxyhemoglobin desaturation curve among freedivers. This is important for understanding the development of hypoxia that could lead to BO. By detecting the S_{pO_2} curve inflection point, it was possible to categorize three different groups of freedivers (*No-desaturation*, *Slow*, and *Fast*). Freedivers that never reached that point during the second apnea had higher S_{pO_2} nadir at the end of the third apnea.

Diving bradycardia is known for having an oxygen-conserving effect (Andersson and Schagatay 1998; Lindholm et al. 1999; de Bruijn et al. 2009; Costalat et al. 2017). In **Study IV**, the initial bradycardia exhibited a comparable magnitude across groups, yet the absolute value was higher in the *Fast* group compared with the other two groups. Moreover, the *Fast* and *Slow* groups showed a further drop in HR in the last section of the 3-minute apnea (A3). This drop in the HR after the initial plateau (Fig. 2) signals the third phase of the diving bradycardia (Sivieri et al. 2015). The physiological mechanism underlying this third phase is not clear. Based on the results of **Study IV**, it can be speculated that in the *Fast* group, the first IBM started around the middle of the third apnea, but the third phase of HR started after 70% of it. This temporal discrepancy suggests that these

two phenomena might not be related. The primary distinctions among the groups were the desaturation rate and the level of oxygen S_pO_2 nadir. The initiation of the third phase of the HR response occurred at 124 seconds, very close to the time to desaturation inflection point (t_{SD}) of the *Fast* group in the last apnea, while S_pO_2 was still at 96%. Traditionally, a S_pO_2 of 95-96% has been considered a breaking point for initiating the third phase of HR kinetics during apnea (Costalat et al. 2017). However, the results from **Study IV** propose that the change in the slope of the desaturation curve could serve as a meaningful breaking point landmark. The oxyhemoglobin dissociation curve undergoes notable alterations in response to varying P_aO_2 levels, mainly manifesting a significant shift within the range of 49 to 59 mmHg (Powers 2016). The third phase of HR modulation during breath-holding is possibly a mechanism to increase oxygen conservation. **Study IV** data strengthen the notion that this phenomenon might be linked to a hypoxic response, especially given its exclusive presence in the groups with more significant desaturation. The results of this study on a large cohort of freedivers affirm that early diving bradycardia is associated with an enhanced oxygen-conserving effect, aligning with earlier findings (Andersson and Schagatay 1998; de Bruijn et al. 2009). On the other hand, freedivers who do not experience early strong bradycardia undergo hypoxemia, and the subsequent decrease in heart rate would not prevent hypoxia but is a consequence of it.

5.1.2 Splenic contraction

The findings from **Study I** confirm that splenic contraction is associated with the degree of hypoxemia, which is in line with previous studies (Richardson et al. 2008). There was a strong positive correlation between the drop in S_pO_2 and the change in Hb, indicating that the spleen contraction could be stronger in the most hypoxic individuals. In **Study I**, the splenic contraction was not influenced by hypercapnia, as the $P_{ET}CO_2$ values were in the range of light hypocapnia.

In **Study I**, the splenic contraction reached 28%, a larger response compared to previous studies that reported a contraction of 16% (Lodin-Sundström and Schagatay 2010) and 18% (Richardson et al. 2008). Those earlier studies employed a less severe hypoxic exposure, precisely 14.2% and 12.8% of inspired oxygen, respectively. The spleen response was faster in **Study I** than reported previously (Lodin-Sundström and Schagatay 2010), probably also related to the higher degree of hypoxic exposure. This observation suggests that splenic contraction is not solely influenced by the extent and duration of exposure but also by the decline in blood oxygenation. This aligns with the earlier proposition that the pace of desaturation during normobaric hypoxia could potentially act as a modulating factor for spleen contraction (Lodin-Sundström and Schagatay 2010). The absence of a correlation between the S_pO_2 nadir and the percentage of

splenic contraction could be attributed to the significant inter-individual variation in both variables. It's important to recognize that hypoxemia alone is not the sole inducer of splenic contraction; it is also linked with the stress response and the release of catecholamines (Stewart et al. 2003; Purdy et al. 2019). The splenic contraction observed in **Study I** was lower than in **Study II**, emphasizing the central role of apnea in modulating splenic contraction. This aligns with recent findings from a study that systematically compared the effects of hypoxia and hypercapnia during normal breathing and breath-holding (Persson et al. 2023), showing that apnea in itself enhanced the contraction. Cumulatively, from the results of **Studies I** and **II**, is it possible to conclude that splenic contraction is probably mediated by a complex interplay involving afferent information from lung stretch receptors sensitive to breath-holding, central and peripheral chemoreceptors responding to hypoxemia and hypercapnia (Richardson et al. 2012; Elia et al. 2022).

5.1.2.1 Splenic contraction and hemoglobin elevation

Hb levels exhibited a post-apnea increase in both **Study I** and **Study IV**, demonstrating similar increments. This is important, considering that the hypoxemia reached during **Study I** was more profound and prolonged, highlighting the effect of apnea on splenic contraction. Although splenic contraction was not explicitly measured in **Study IV**, the observed rise in Hb percentage aligns comparably with the

results of **Study I** and other investigations utilizing hypoxia as the experimental variable (Richardson et al. 2008; Schagatay et al. 2020). Then, the baseline spleen volume and the magnitude of splenic contraction are important for maintaining CaO_2 during breath-holding and thus reducing the possibility of brain hypoxia.

It is important to note that a substantial individual variation in the Hb response is observed in both **Study I** and **Study IV**, which could explain the lack of correlation between splenic contraction and Hb elevation in **Study I**. Nevertheless, both studies showed an association between the decline in $\text{S}_\text{p}\text{O}_2$ and the Hb response. This suggests that the heightened sympathetic activity, triggered by peripheral receptors sensing hypoxemia, could stimulate splenic contraction, which in turn might be connected to the subsequent elevation in Hb with the degree of hypoxemia. In fact, the Hb levels did not show a significant elevation in the *No-desaturation* group in **Study IV**, only in the other two groups, highlighting the effect of hypoxia on the splenic contraction, which is a mechanism that protects the freediver when hypoxemia arises.

5.1.2.2 Desaturation rate and spleen volume

Baseline spleen volume was higher in the *No-desaturation* group than in the other two. This finding from **Study IV** ratifies the importance of the spleen in breath-hold diving activities. The spleen serves as a blood reservoir during breath-hold diving (Espersen et al. 2002), with

its contraction enhancing CaO_2 . Consequently, it's logical to infer that freedivers with larger spleens would experience slower desaturation, as observed in the *No-desaturation* group. Although contraction was not measured in **Study IV**, the resting splenic volume is positively correlated with the magnitude of splenic contraction (Schagatay et al. 2015; Holmström et al. 2020). Additionally, research suggests that dynamic apnea (Yang et al. 2022) and endurance training (Holmström et al. 2021) can increase resting splenic volume. As mentioned before, larger spleen volume could be associated with the training intensity. Moreover, the Bajau people, renowned for their breath-holding abilities, may have enlarged spleens due to a genetic adaptation (Ilardo et al. 2018). This raises the question of whether larger spleen volume is the result of training or natural selection in competitive breath-hold divers. Individual spleen size and function are likely partially genetically determined and partially trainable.

5.2 Oxygen stores and oxygen consumption

5.2.1.1 Desaturation rate as a surrogate of oxygen consumption

During equal-duration apneas, a stable oxyhemoglobin dissociation curve signifies an optimal alignment of factors influencing arterial oxygen content, including the body's oxygen stores, lung oxygen diffusion, and ventilation-perfusion ratios, with $\dot{V}\text{O}_2$. Consequently, the onset of desaturation serves as a signal that P_aO_2 is dropping along with CaO_2 . In the initial part of a static apnea, the lungs serve as

the primary reservoir for oxygen (Rahn 1964), and the diffusion process relies on the pressure gradient of oxygen between the lungs and the capillaries. The reduction in $P_{A}O_2$ follows a non-linear decline (Taboni et al. 2020), so the reduction is slow at the beginning of apnea and becomes faster towards the end. At low $P_{A}O_2$, the blood oxygen stores become the main oxygen reserve (Rahn 1964). This is one of the reasons why comparing oxygenation variables between different duration apneas is difficult. As the breath-hold progresses, the alveolar volume is reduced as oxygen is consumed, and the reduction in $P_{A}O_2$ would lead to a decrease in diffusion. Blood perfusion could also be reduced due to hypoxic pulmonary vasoconstriction (Moudgil et al. 2005). Furthermore, the effects of hypoxemia on shunt quantity will affect the $P_{A}O_2$ along the breath-hold (Rodríguez-Roisin and Roca 2005).

When comparing the estimated oxygen stores of participants in **Study IV** between the *No-desaturation* and *Fast* groups, the difference is minimal. Since lung packing was not performed, it is feasible to utilize VC to calculate lung oxygen stores. Calculating the $C_{a}O_2$ involves using Hb values, baseline S_pO_2 , and assuming a $P_{A}O_2$ of 100 mmHg. As the mean body mass was close to 70 kg, a blood volume of 5 L can be approximated (Feldschuh and Enson 1977). The estimated oxygen stores in the *No-desaturation* group amount to ~2067 ml, while in the *Fast* group, they are ~2009 ml. The contribution of splenic contraction can be disregarded as the splenic contraction was not

measured in this study, and Hb did not increase in the *No-desaturation* group. Nevertheless, given the relatively small differences in oxygen stores, the actual difference between groups could depend on variations in $\dot{V}O_2$. The observed differences in the HR also support this conclusion, as the *No-desaturation* group had a lower HR than the *Fast* group in baseline and nadir levels, which aligns with findings in other diving mammals (Williams et al. 1991; Hindell and Lea 1998; Webb et al. 1998).

The importance of identifying the onset of desaturation ($tS\Delta$) is that it indicates when the body can no longer maintain the CaO_2 at the same values as baseline. Since no individual changes in oxygen stores were anticipated during the protocol in **Study IV**, the timing of desaturation initiation during the apnea could indicate differences in $\dot{V}O_2$. The $tS\Delta$ was delayed during the last apnea compared with the second, a finding that could indicate lower $\dot{V}O_2$ in A3. PAO_2 and P_aO_2 are challenging to measure outside specialized laboratories but using the $tS\Delta$ to determine when oxygen desaturation starts is a practical way to address an imbalance between oxygen stores, oxygen transfer, and $\dot{V}O_2$.

5.2.1.2 Desaturation rate and vital capacity

In **Study IV**, no significant differences were observed among the groups regarding VC. This finding was surprising as the lungs are an important oxygen store, and larger lung volume has been associated

with a slower desaturation rate (Findley et al. 1983) and higher end-apnea P_{AO_2} (Hong et al. 1971). Additionally, larger lung volumes were associated with increased apneic performance (Schagatay et al. 2012). However, it's essential to recognize that the studies of Findley (1983) and Hong (1971) are not directly comparable to **Study IV** findings, as they compared TLC against functional residual capacity and residual volume, and all our participants were holding their breath close to the VC. Additionally, it's important to note that **Study IV** focused on competitive freedivers performing a sub-maximal static apnea, so the oxygen stores were not used to the limit, as in a maximal duration static apnea. **Study IV** participants also exhibited vital capacities exceeding predicted values (Quanjer et al. 2012), with males reaching 130% and females at 122% of expected values. To conclude, these results do not contradict previous studies but suggest that in a trained population, the main determinant of the oxygen desaturation rate during sub-maximal static apnea is not the VC.

5.2.1.3 Desaturation rate and BMI

In **Study IV**, the BMI was lowest in the *No-desaturation* group. It is important to note that having a lower BMI does not necessarily imply that a person has a lower percentage of body fat (Flegal et al. 2009). Additionally, hypoxic training has been associated with body fat reduction (Bonne et al. 2023; Urdampilleta et al. 2023; He et al. 2023). So, it is difficult to separate the effects of hypoxic training in the

protective mechanisms from the reduction in body mass. In other words, a freediver with high-intensity training could have improvements in the diving response (Schagatay et al. 2000) and spleen volume (Bouten et al. 2019) along with loss of body mass, making it difficult to interpret BMI isolated from the other variables. In fact, the static apnea personal best in the *No-desaturation* group was also the longest, suggesting that this group could be more trained in this discipline. In **Study IV**, no data were collected on changes in body mass within the previous 24 hours. This information is crucial as acute weight loss has been reported to reduce the basal metabolic rate (Dulloo and Jacquet 1998). To conclude, caution is warranted when interpreting the lower BMI observed in the *No-desaturation* group. This may be due to several factors that have different effects on the desaturation rate, and these should be studied separately.

5.3 Risk factors for hypoxemia.

5.3.1 Hyperventilation

Hyperventilation is considered to increase the risk of BO because it delays the physiological breaking point and is associated with lower S_pO_2 and $P_{ET}O_2$ at the end of apnea, hence increasing the risk of severe hypoxia (Craig 1961, 1976; Lanphier and Rahn 1963; Hill 1973).

Hyperventilation reduces the central stimulus as P_aCO_2 is lower than normal at the start of the breath-hold (Caruana-Montaldo et al. 2000).

Hyperventilation also increases V_t and cyclically unloads stretch

receptors, which typically will have negative feedback on the respiratory drive. The interaction of multiple factors like chemoreceptors and lung stretch receptors stimulation, neural drive, and cognitive factors determines the apnea breaking point. Hyperventilation delays the physiological breaking point, subsequently increasing the duration of the apnea. Data from **Study II** strengthened the prominent role of $P_a\text{CO}_2$ in determining the physiological and absolute apnea breaking points. In the hyperventilation trial, participants exhibited lower $P_{\text{ET}}\text{CO}_2$, $P_{\text{ET}}\text{O}_2$, and $\text{S}_\text{p}\text{O}_2$ compared to the normal breathing trial. Despite the more significant hypoxemia during hyperventilation, IBM were delayed, and the apnea duration was prolonged, indicating that hypoxemia may not reliably serve as an intrinsic warning signal. When comparing the initiation of IBM in **Study II** with **Study IV**, it was observed that the onset time of IBM during the hyperventilation trial (112 seconds) was akin to the time noted in **Study IV**. Thus, the physiological breaking point in non-divers following hyperventilation resembles that of trained freedivers, albeit reaching a lower level of $\text{S}_\text{p}\text{O}_2$.

5.3.1.1 Hyperventilation and oxygen desaturation

Study II revealed the novel finding that hyperventilation led to progressive desaturation in a series of dives. Several factors could explain this serial desaturation. Firstly, the prolonged apnea

duration, as explained in the previous paragraph. Secondly, voluntary hyperventilation can increase the work of breathing (Otis 1954). This implies initiating the breath-hold with an elevated $\dot{V}O_2$. In a series of apneas, starting with an increased $\dot{V}O_2$ could lead to a faster desaturation in mixed venous blood that does not return to baseline values, if the surface interval is not long enough to “pay” the oxygen debt. Consequently, the following apnea will start with mixed venous hypoxemia and an increased pulmonary oxygen uptake that would drain the oxygen stores quicker during the dive (Sands et al. 2010).

Despite initiating a breath hold dive with higher $P_{ET}O_2$ values in the hyperventilation condition, the $P_{ET}O_2$ values were lower at the end of apnea compared to normal breathing. Additionally, even with the rise in P_AO_2 induced by hyperventilation and its influence on P_aO_2 , the S_pO_2 nadir was lower during the hyperventilation trial. Moreover, the S_pO_2 maintained a consistent value when comparing apneas of the same duration. These findings emphasize the absence of benefits from hyperventilation, as there is no improvement in the oxygenation variables. However, they also highlight the potential risk linked to increased exertion on respiratory muscles, which may lead to elevated $\dot{V}O_2$.

5.3.1.2 Hyperventilation and the diving response

Study II is the first study to explain how hyperventilation affects the diving response. Despite similar baseline HR between conditions, during the first 30 sec of the breath-hold, the participants in the hyperventilation trial showed a higher HR than during the normal breathing trial. The mean HR during apnea was similar between trials. Different factors could cause the increase in HR at the beginning of apnea. Hyperventilation increases HR via the autonomous nervous system, whether it is parasympathetic attenuation or sympathetic activation (Hawkins et al. 2019). Additionally, hyperventilation increases stroke volume due to increases in the cardiac preload as the swings in intra-pleural pressure have a pumping effect (Cummin et al. 1986). Lastly, the increased work of breathing due to hyperventilation can lead to increased $\dot{V}O_2$, which also increases the HR (Coast and Krause 1993).

The second component of the diving response, peripheral vasoconstriction, estimated via blood flow in the hand, did not show differences between conditions. However, we found high differences among participants, indicating that the sympathetic response to apnea is highly individual (Busch et al. 2019).

Even though the S_pO_2 nadir was lower in the hyperventilation trial, the strength of the diving response was the same as normal breathing. Thus, no compensation occurred, allowing longer apneas

without greater oxygen conservation. This once again demonstrates that there is no real advantage to hyperventilation, only an increased risk of hypoxic BO.

5.3.1.3 Hyperventilation and splenic contraction

The reduction in spleen volume during apnea in **Study II** was meaningful in both conditions, and the spleen did not revert to its baseline volume between consecutive apneas. However, no significant differences were observed in spleen volumes between normal breathing and hyperventilation or across the series of apneas. This indicates that this protective mechanism did not counteract the increased desaturation observed with prolonged apneic duration. It is worth noting that in prior studies, spleen volumes were exclusively measured between apneas (Hurford et al. 1990; Espersen et al. 2002; Baković et al. 2003; Schagatay et al. 2005; Richardson et al. 2012; Elia et al. 2021a) or during short-duration apneas (Palada et al. 2007). In contrast, **Study II** involved measurements along all the protocol and with more prolonged duration apneas, showing that the spleen reduces its volume to 35% during the breath-hold. Remarkably, splenic contraction during apnea, which is enhanced by hypoxia (as shown in **Study I**) and probably by hypercapnia (Richardson et al. 2012), remained similar in both conditions. This finding may be due to a dynamic balance between increased desaturation and reduced hypercapnia during hyperventilation across the apnea series,

suggesting that both stimuli could be important for the magnitude of the contraction.

5.3.1.4 Hyperventilation and blackout

In **Study III**, it was found that divers tend to hyperventilate by increasing V_t . Additionally, from **Study II**, despite increased hypoxemia, hyperventilation does not modify the diving response or the splenic contraction. Hence, hyperventilation increases the risk without enhancing the protective factors. Despite hyperventilation's documented reduction in brain blood flow (Viola et al. 2012), breath-holding induces increased flows in both cerebral (Kjeld et al. 2009) and carotid (Joulia et al. 2009) circulations. Still, during breath-holding, brain tissue oxygenation remains stable (Ratmanova et al. 2016; Vestergaard and Larsson 2019). To date, there is no information on the effects of hyperventilation before breath-holding on brain tissue oxygenation, which is why our group started a pilot study to compare brain oxygenation via near-infrared spectroscopy between apneas preceded by normal breathing and by hyperventilation. Moreover, it is noteworthy that in the series of simulated dives in **Study II**, the magnitude of the diving bradycardia and spleen contraction remained consistent across the series. In conclusion, during serial diving, there is no enhancement of protective mechanisms with subsequent dives.

5.4 Sex differences

All studies in this thesis included both male and female participants. In each study, except for **Study III**, an analysis was conducted to assess potential differences between the sexes. Although the aim of this thesis is not to account for sex differences, it was essential to establish if the sexes affected the results.

Upon collectively examining the data from **Studies I, II, and IV**, it became evident that males, on average, exhibited higher vital capacity and larger spleen volume, not only in absolute terms but also when expressed in proportion to body size. In **Study I**, there were no differences between the sexes in Hb. However, in **Study IV**, which had a larger sample size, males indeed demonstrated higher Hb levels than females. These findings align with population surveys (Quanjer et al. 2012; Murphy 2014; Chow et al. 2016), showing that those differences are expected in the general population.

The analysis of the diving response in **Studies II and IV** did not reveal any differences between sexes in the magnitude of diving bradycardia or vasoconstriction, aligning with previous results (Peng et al. 2021). Additionally, in **Study II**, no distinctions were observed between the sexes on the diving response magnitude during hyperventilation. In **Studies I and II**, despite females' smaller baseline spleen volume, the magnitude of splenic contraction was similar between sexes.

In **Study IV**, no significant disparity in the oxygen desaturation profile was observed between the sexes. Although males exhibited elevated oxygen stores characterized by a larger VC and higher Hb levels alongside a larger spleen, those higher oxygen stores did not affect the oxygen desaturation rate in this sub-maximal static apnea test.

5.5 Practical applications

How can the results from all the studies be applied to avoid hypoxic BO? Hyperventilation should be avoided. But from the results of **Studies II, III, and IV**, it doesn't seem easy even after the explicit instruction not to do it. Additionally, an increased spleen volume and a strong diving response are beneficial to avoid severe hypoxemia, as showed in **Studies I and IV**, and they seem to be trainable. Finally, the ramp apnea test could be an approach to estimate individual $\dot{V}O_2$ and to identify who is at higher risk of fast oxygen desaturation. I will develop these ideas further.

5.5.1 Avoiding hyperventilation

Maintaining a normal breathing pattern before holding one's breath proved to be challenging, both for freedivers and non-freedivers.

During **Study II**, even during the normal breathing trial, most of the participants performed mild hyperventilation, and during the hyperventilation trial, the magnitude of this breathing pattern

increased along the series and was associated with more severe desaturation. During **Studies III** and **IV**, the participants were instructed to avoid hyperventilation; however, most of them could not avoid it. On the other hand, the lowest $P_{ET}CO_2$ values when comparing **Studies II, III**, and **IV** were during the hyperventilation trial in **Study II**; this suggests that at least trying to avoid it could reduce its magnitude. Therefore, it should be emphasized in instructions to divers that it is essential to minimize hyperventilation.

Study III showed that estimating hyperventilation before breath-holding was possible, and it could be used in an underwater environment. The predicted $P_{ET}CO_2$ could be used as a surrogate of actual measurement. During the apnea ramp test, it was observed that freedivers tend to hyperventilate by increasing V_t instead of respiratory rate. This is a critical point as the respiratory rate is easy to measure just by simple observation, but V_t is more difficult to estimate. Thus, just counting the respiratory rate is not enough to assess hyperventilation in freedivers. Furthermore, it opens the possibility to estimate the degree of hyperventilation, a matter of significance, as pronounced hyperventilation can not only prolong dives but also lead to cerebral and cardiac ischemia (Laffey and Kavanagh 2002; Rutherford et al. 2005). Indeed, according to the results of **Study II**, the magnitude of the hyperventilation makes a difference. The implications of this exploratory study could lay the foundation for the development of a wearable system capable of

monitoring freedivers during their actual dives, as S_pO_2 could also be monitored underwater (Mulder et al. 2021). By combining the data on oxygen desaturation with information on respiratory patterns, redundant information could be obtained to alert the diver about an increased risk of BO during serial freediving.

5.5.2 Static apnea ramp test

In **Study IV**, a static apnea ramp test was found to be an effective tool for detecting freedivers at risk of faster oxygen desaturation. During the second apnea, monitoring S_pO_2 behavior could indirectly evaluate $\dot{V}O_2$. The static apnea ramp test is also useful for assessing the magnitude of bradycardia in a standardized way, as improvements in this response could translate into improved performance (Schagatay 2009). Regularly conducting the test could also provide information on changes in the diving response and the body's tolerance to increased levels of P_aCO_2 by measuring the time for the first IBM, allowing athletes and coaches to evaluate objectively the result of training interventions. However, further research is needed to determine whether the test's results can be replicated in a dry environment. If we simplify the interpretation of the S_pO_2 profile, we can categorize individuals as follows: If there is desaturation within a 2-minute apnea, they are in the *Fast* group; if there is no desaturation at any moment, they are in the *No-desaturation* group; and if desaturation happens after the apnea, they are in the *Slow* group. This

categorization could simplify the implementation of the apnea ramp test without requiring complex computational methods.

5.5.3 Apnea training

The average static apnea personal best in the *No-desaturation* group was close to 6 minutes, while in the other two groups, it was around 5 minutes. This raises the question of whether training is responsible for an increased spleen and a stronger diving response. Previous research has shown that endurance (Holmström et al. 2021) and apneic training (Bouten et al. 2019, Yang et al. 2022) can increase the spleen volume, although training duration could affect this response (Elia et al. 2021b). The diving response is enhanced by breath-hold training (Engan et al. 2013). Additionally, trained freedivers show an enhanced diving response (Schagatay et al. 2000, Delapille et al. 2002, Lemaître et al. 2013) and larger spleen size (Schagatay et al. 2012). Adequately supervised breath-hold training could, therefore, benefit individual safety during apnea while underwater, such as spearfishers and snorkelers.

6 Limitations

Peripheral oxygen saturation

During **Studies I, II, and IV**, the analysis of the desaturation profile included change detection as well as specific values. Apnea induces vasoconstriction, which could affect signal quality. Therefore, throughout all measurements, signal quality was meticulously assessed, ensuring the exclusion of values that could be attributed to poor signal acquisition. Additionally, given that the oxyhemoglobin dissociation curve shifts to the right with increased $[H^+]$ and P_aCO_2 , prolonged apnea may lead to an underestimation of P_aO_2 by S_pO_2 . In **Study IV**, minimal lactate accumulation was observed. Considering the $P_{ET}CO_2$ at the end of apnea, the calculated pH would be 7.26 ± 0.05 , suggesting a limited impact on the oxyhemoglobin dissociation curve.

Hyperventilation protocol

In **Study II**, V_t lacked specific control, as the breathing pattern adhered to instructions provided by a researcher. The hyperventilation trial aimed to escalate both respiratory rate and V_t , whereas in the normal breathing trial, participants were directed to breathe naturally. The results of **Study II** should be interpreted in the context of severe hyperventilation, with the normal trial representing a milder form of hyperventilation. Since the participants lacked prior

breath-holding experience, maintaining their usual breathing pattern after the command to breathe normally was anticipated. While this may be perceived as a variable control limitation, replicating a prescribed breathing pattern for respiratory rate and V_t control is challenging in real-world breath-hold diving. Thus, the study results gain ecological validity under conditions where such control is impractical.

End-tidal CO_2 estimation via a force sensor

Females are underrepresented in **Study III**, which affects its applicability to the overall population. Ideally, measurement should have included the VC and V_t during the apnea test. Still, the purpose of the field laboratory was to keep it simple enough to avoid affecting the athletes' already tight schedules. The accuracy needs to be improved, and it should be evaluated in actual diving conditions to ensure its effectiveness.

7 Conclusions

7.1 Main conclusions

This thesis has identified hyperventilation before serial freediving as a risk factor associated with developing severe hypoxia and the risk of BO. It has further studied some protective mechanisms: spleen blood boosting and the cardiovascular diving response. However, these mechanisms do not compensate for the development of an oxygen debt in serial diving. A static apnea ramp test was developed to identify individuals at increased risk for hypoxic BO, with the diving response and spleen volume emerging as personal distinctive protective factors. A wearable device for identifying hyperventilation from chest movements was developed and tested, with encouraging results. This is a proof of concept, and further development is required to increase its accuracy. The main conclusions of the four studies in this thesis are addressed point by point below.

Study I revealed that acute eupneic hypoxia, as short as 3 min, caused significant splenic contraction. This suggests that a long single apnea would induce acute splenic contraction and an increase in hemoglobin, which protects the freediver against hypoxia.

In *Study II*, spleen function was studied in a series of 5 consecutive apneas, and its magnitude of contraction was found to be similar in all the dives: The protective effect persisted uniformly across all dives

but was insufficient to offset increased desaturation during prolonged apneas. The spleen contracted the most during the first breath hold. Therefore, freedivers do not have an increased risk on the first dive of a series.

Study II further revealed the novel finding that hyperventilation before breath-holding caused increasing desaturation during five simulated serial dives, a feature that was not observed with normal breathing. This observation suggests the possibility of accumulating an oxygen debt due to extended apneas in the hyperventilation trial. Additionally, it raises the question of whether increased $\dot{V}O_2$ resulting from increased respiratory effort could be a contributing factor.

Study III revealed that freedivers hyperventilate without noticing it, even when instructed to avoid it. They increased their V_t while maintaining a normal respiratory rate. This finding was confirmed by the results from *Study IV*, where exhaled end-tidal CO_2 decreased from A1 to A3.

Study III is a first step towards addressing hyperventilation in real-world freediving scenarios, paving the way for the development of a monitoring system to alert the diver of hyperventilation.

Study IV showed for the first time how a newly developed simple apnea ramp test could identify freedivers more prone to desaturation.

These vulnerable divers are desaturated, already in a 2-minute static apnea, and it would be expected that they would be at risk of pronounced hypoxia in actual dives. Moreover, divers prone to desaturation after 2 minutes typically exhibit a less prominent diving response and smaller spleens compared to those who do not experience desaturation. These at-risk freedivers probably have higher oxygen consumption and smaller body oxygen stores. A pre-diving static apnea ramp test could be helpful to set individual limits in a future warning device.

Overall, the studies that comprise this thesis added helpful information on the complex mechanisms of hypoxia development during breath-hold diving. It provides a promising basis for mitigating behavior associated with hypoxic BO and for developing a freediver-warning system for the detection of potentially dangerous hypoxia. The present work also identifies some new questions that would require further research.

7.2 Secondary conclusions

In **Studies II** and **IV**, it was found that the diving response and splenic contraction display uniformity across the sexes. Despite females having smaller oxygen reserves, the level of hypoxemia was comparable to that of males in both studies. The consistent duration of apneas in **Study IV** ratifies that the balance between oxygen stores and oxygen consumption is the same between the sexes.

Study I confirmed the strong correlation between baseline splenic volume and the magnitude of splenic contraction. *Study IV* found that freedivers with larger spleens exhibited higher S_pO_2 levels at the end of apnea, which could indicate the effects of improved diving response and splenic volume with apneic training.

Based on the findings from *Study IV*, the second drop in HR, indicative of the third phase of diving bradycardia, is likely an acute response to accentuated hypoxemia, which enhances the oxygen-conserving mechanism.

8 Future directions

Hyperventilation effects during more extended series of apneas require further study to understand better how diving and resting intervals should be paced. Recreational breath-hold divers face a particular risk of experiencing hypoxic BO, mainly due to their exposure to serial apneas. This exploration will help determine whether there is a ceiling effect in the level of hypoxemia or if a longer series further escalates the risk. Serial diving must also be studied during actual dives without intentional hyperventilation, as divers attempting to avoid hyperventilation still seem to do it anyway. It should further be determined how long a pause in serial diving is needed to recover fully before diving can be resumed in resting and working conditions. Diving pacing strategies should also be further studied by serial freediving experts among professional harvesting divers, e.g., the Ama of Japan and the Bajau of Indonesia, to understand better how they avoid BO.

A model to predict individual oxygen desaturation could be developed based on the information from the static apnea ramp test and the exploration of training intensity and characteristics to account for the differences in diving response and spleen volume. This should follow the development of a dynamic apnea ramp test to incorporate the effects on underwater swimming. Finally, we should

explore how to apply this to real diving situations. Pilot testing of these strategies is ongoing.

Brain oxygenation during hyperventilation needs further study to gain a complete understanding of the development of hypoxic BO. As hypoxemia escalates during apneas following hyperventilation, understanding whether hyperventilation reduces blood flow to the brain becomes crucial. A static apnea ramp protocol, contrasting normal breathing with hyperventilation, could prove instrumental in discerning these distinctions. Furthermore, since estimating brain tissue oxygenation and blood flow can be achieved non-invasively, it lends itself well to large-scale study designs.

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